

The Symmetric Analogue of the Polynomial Power Basis

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A new polynomial basis over the unit interval $t \in [0, 1]$ is proposed. The work is motivated by the fact that the monomial (power) form is not suitable in CAGD, as it suffers from serious numerical problems, and the monomial coefficients have no geometric meaning. The new form is the symmetric analogue of the power form, because it can be regarded as an “Hermite two-point expansion” instead of a Taylor expansion. This form enjoys good numerical properties and admits a Horner-like evaluation algorithm that is almost as fast as that of the power form. In addition, the symmetric power coefficients convey a geometric meaning, and therefore they can be used as shape handles. A polynomial expressed in the symmetric power basis is decomposed into linear, cubic, quintic, and successive components. In consequence, this basis is better suited to handle polynomials of different degrees than the Bernstein basis, and those algorithms involving degree operations have extremely simple formulations. The minimum degree of a polynomial is immediately obtained by inspecting its coefficients. Degree reduction of a curve or surface reduces to dropping the desired high degree terms.

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1. INTRODUCTION

1.1 The Bernstein Versus the Power Form

The Bernstein-Bézier curve or surface form has become a de facto standard in a CAGD context, as it enjoys many advantages [Farin 1993; Bowyer and

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Woodwark 1993]. First, it exhibits elegant geometric properties, including the invariance under affine transformations, containment in the convex hull, and an intuitive geometric interpretation of the coefficients as control points. Second, it is very stable from a numerical viewpoint [Farouki and Rajan 1987]. Notwithstanding these positive properties, the Bernstein form has also some weak points:

- The actual or minimum degree $n_{\min}$ of a polynomial in Bernstein form is not immediately apparent by inspection of its coefficients, but requires a complex procedure [Farouki and Rajan 1988].
- The degree reduction algorithm tends to be numerically unstable, as pointed out by Goodman and Said [1991b], because the calculation of the new Bézier points involves nonconvex combinations of the original ones.
- This form is overly complicated for representing polynomials of different degrees in a homogeneous way, because they all must be raised to a common degree, the highest one. Furthermore, although degree elevation is stable, it may produce roundoff.

In short, the Bernstein form is not very efficient when degree modification is involved. This is the case of certain algorithms for designing surfaces from curve input, where the degrees of these curves must be modified until they reach a common degree.

In contrast, the monomial basis $\{1, t, t^2, \dots, t^n\}$, the so-called power basis, is well suited to handle polynomials with different degrees, as there is no need of conversion to a common degree. In this basis the actual or minimum degree $n_{\min}$ of a polynomial is just the subscript of the last nonzero coefficient. In addition, the monomial form is efficiently evaluated using Horner's method (also known as nested multiplication), which, for a degree- n polynomial, requires only n multiplications and n additions. If the degree of a polynomial reduces, this information immediately speeds up evaluation in the power form, whereas in the Bézier case a new set of polynomial coefficients must be computed. In addition, the usual de Casteljau evaluation algorithm for the Bernstein form is relatively expensive, as it shows an $O(n^2)$ growth in running time. Although an $O(n)$ nested form can be devised for the Bernstein case [Farin 1993], the power form is still faster.

A software developer might jump to the conclusion that, given a polynomial in Bernstein form, it should be transformed to the monomial form in those algorithms that are more efficient in the latter. Nonetheless, there are some fundamental arguments against this strategy:

- (1) The transformation matrices between the Bernstein and power forms are ill-conditioned; in other words, the basis conversion implies a severe loss of accuracy.
- (2) The monomial form shows poor numerical condition. Slight perturbations in the coefficients of a polynomial in the monomial form affect it much more than in the Bernstein form [Farin 1993].

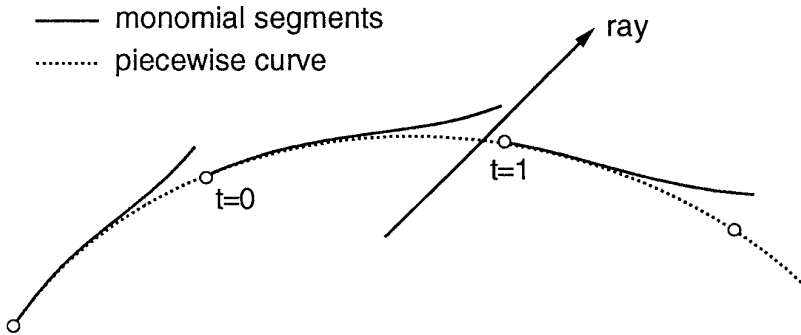


Fig. 1. Lack of reliability near $t = 1$ for the monomial form.

The power form enjoys good numerical properties near $t = 0$, but it deteriorates rapidly when approaching $t = 1$. Such lack of reliability near $t = 1$ leads to gaps between adjacent curve segments or surface patches, especially for high degrees. This phenomenon may not be noticeable when rendering curves or surfaces of moderate degree, but even for such low degrees it can cause a system crash when carrying out geometry processing. For instance, in a B-rep solid-modeling system, the set membership classification implies calculating ray-surface intersections [Miller 1986]. This intersection may be missed through these discontinuities, which would result in an invalid topology. The problem is illustrated in Figure 1 for the 2D case. Similarly, dropping the monomial terms of highest degree yields a good approximation near $t = 0$, but it is unacceptable near $t = 1$ because it leads to gaps in the piecewise case. All those deficiencies stem from the fact that the monomial form is essentially a Taylor expansion around $t = 0$. It is good at representing functions over a symmetric interval around $t = 0$, but piecing together several Taylor expansions can be considered a very “unnatural” scheme.

1.2 The Symmetric Analogue of the Power Form

This article presents a new polynomial basis over the unit interval $t \in [0, 1]$ that can be regarded as the proper symmetric analogue of the power basis because it is derived by imposing basic requirements:

- (1) Symmetry, in that replacing the variable t by $(1 - t)$ results in the same set of basis functions.
- (2) The basis functions of low degree are a subset of those for high degree, so that a polynomial in the basis is expanded into terms of increasing degrees.

We can think of the resulting polynomial form as an “Hermite two-point expansion,” because a truncation results in an Hermite interpolant defined by successive derivatives of the original polynomial at $t = 0$, $t = 1$, instead of a Taylor expansion. Thus this new form is suitable for the definition of

piecewise curves or surfaces. It preserves the advantages of the power form over the Bernstein form and remedies the deficiencies of the former:

- It can be evaluated using a Horner-like scheme almost as fast as that of the power form.
- The minimum degree of a polynomial is immediately obtained from its coefficients.
- It enjoys numerical stability near both $t = 0$ and $t = 1$, thereby avoiding undesirable gaps in the piecewise case.
- The conversion matrices between the Bernstein and symmetric power forms are not ill-conditioned.
- The symmetric power coefficients do admit a geometrical interpretation, related to the inner Bézier points of the aforementioned Hermite interpolants.

The article is arranged as follows. Section 2 defines the symmetric power form of a polynomial. In Section 3, explicit expressions of the conversion matrices between the Bernstein and symmetric power forms are obtained, and related numerical issues are studied. Section 4 analyzes the numerical properties of the symmetric power form through condition numbers. In Section 5 the geometric interpretation of the symmetric power coefficients is obtained, and results for curves and surfaces are derived. Finally, conclusions are drawn in Section 6.

2. DEFINING THE SYMMETRIC POWER FORM

2.1 Symmetry Condition for the Basis Functions

As already stated in the Introduction, the goal is to obtain a polynomial basis over the unit interval $t \in [0, 1]$ that is the symmetric analogue of the power basis. The symmetry condition imposes that, replacing the variable t by $(1 - t)$, keeping $[0, 1]$ as the domain, must result in the same set of basis functions. A candidate basis can be constructed by choosing a suitable subset in the set of polynomials taking the form:

$$\tilde{P}_{i,j}(t) = t^i(1 - t)^j. \quad (1)$$

Such nonnegative polynomials can be represented as points in a discrete space of two dimensions corresponding to the exponents i, j , as displayed in Figure 2. The symmetry condition imposes that, if a certain function $\tilde{P}_{i,j}(t)$ belongs to the subset, then $\tilde{P}_{j,i}(t)$ must belong to it too. Equivalently, the points representing the basis functions must be arranged symmetrically with respect to the symmetry axis traced in Figure 2. This figure also shows three degree- n polynomial bases that correspond to very simple patterns:

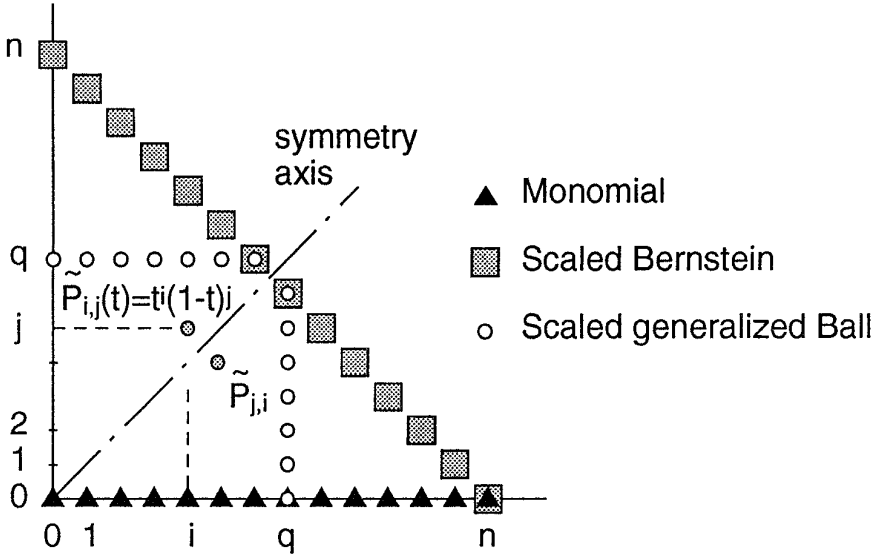


Fig. 2. Graphical representation of polynomials $\tilde{P}_{i,j}(t)$.

—Monomial basis:

$$\tilde{P}_{i,0}(t) = t^i, \quad i = 0, \dots, n. \quad (2)$$

—Scaled Bernstein basis $\tilde{B}_i^n(t)$, obtained after dropping the binomial coefficients in the standard Bernstein polynomials $B_i^n(t)$:

$$\tilde{B}_i^n(t) = \tilde{P}_{i,n-i}(t) = t^i(1-t)^{n-i} \quad i = 0, \dots, n. \quad (3)$$

—Scaled generalized Ball basis (odd degree $n = 2q - 1$) $q \geq 1$, derived in a similar way from the generalized Ball basis [Goodman and Said 1991a]:

$$\left. \begin{aligned} \tilde{P}_{i,n-i}(t) &= t^i(1-t)^q \\ \tilde{P}_{n-i,i}(t) &= t^q(1-t)^i \end{aligned} \right\} i = 0, \dots, q-1. \quad (4)$$

As clearly observed in Figure 2, both the scaled Bernstein and generalized Ball basis fulfill the symmetry requirement, whereas the monomial basis does not.

The degree of a polynomial $\tilde{P}_{i,j}(t)$ can be interpreted as the distance $n = (i + j)$ to the origin. Hence the locus of points $\tilde{P}_{i,j}(t)$ with common degree n is traced out by the scaled Bernstein polynomials of degree n . Observe that, for the Bernstein or Ball basis, increasing the degree implies complete replacement of the original basis functions by new ones, whereas in the monomial case new basis functions are added, preserving the existing ones.

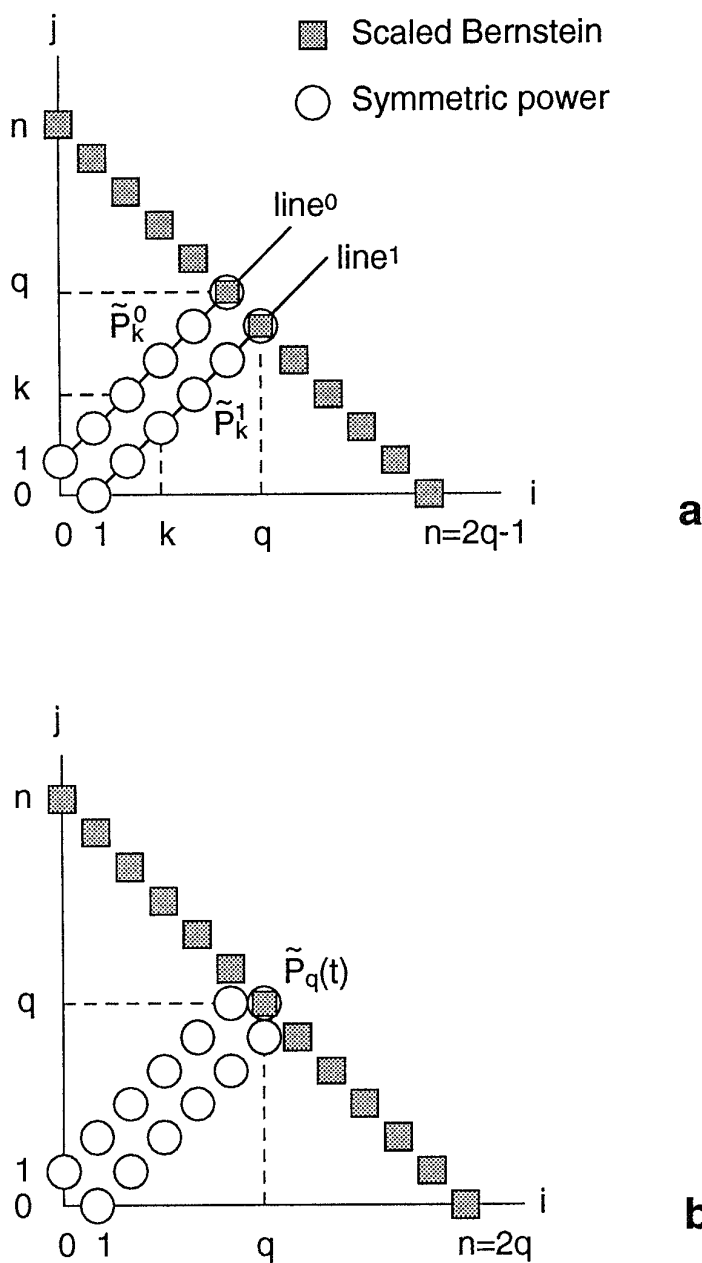


Fig. 3. Symmetric power basis of degree: (a) odd $n = 2q - 1$; (b) even $n = 2q$.

2.2 The Symmetric Power Basis

The symmetric analogue of the power basis must be constructed in a similar way to the latter, that is, adding new basis functions as the degree increases. The obvious way to achieve this goal and simultaneously meet the symmetry condition is to take polynomials $\tilde{P}_k^0(t)$, $\tilde{P}_k^1(t)$ arranged along

two parallel lines 0, 1 at both sides of the symmetry diagonal (Figure 3a). Given an integer $q \geq 1$, the basis functions for the odd-degree $n = 2q - 1$ are:

$$\left. \begin{aligned} \tilde{P}_k^0(t) &= \tilde{P}_{k,k+1}(t) = t^k(1-t)^{k+1} \\ \tilde{P}_k^1(t) &= \tilde{P}_{k+1,k}(t) = t^{k+1}(1-t)^k \end{aligned} \right\} \quad k = 0, \dots, q-1. \quad (5)$$

As in the generalized Ball case (4), we could restrict ourselves to the odd-degree case. Nevertheless, it takes very little effort to define a family of basis functions for the even-degree $n = 2q$ (Figure 3b), by adding to the set (5) the polynomial $\tilde{P}_q(t)$ on the symmetry axis:

$$\tilde{P}_q(t) = \tilde{P}_{q,q}(t) = t^q(1-t)^q. \quad (6)$$

However, note that for $n = 2q + 1$ this polynomial is lost and replaced by two new ones.

The preceding polynomials (5) and (6) are referred to as the symmetric power polynomials. For n both odd or even, there are $n + 1$ polynomials in total. It is easily demonstrated that they form a basis of the vector space (over the real field) of polynomials up to degree n . As this space has dimension $n + 1$, the demonstration reduces to prove that the symmetric power polynomials are linearly independent. This is trivial, because a given pair $\tilde{P}_k^0(t), \tilde{P}_k^1(t)$ is composed of two linearly independent polynomials, and the successive pairs and $\tilde{P}_q(t)$ have increasing degrees, thereby being linearly independent.

A polynomial $p(t)$ is expressed in the symmetric power form as

$$p(t) = \sum_{k=0}^{q-1} [\tilde{a}_k^0 \tilde{P}_k^0(t) + \tilde{a}_k^1 \tilde{P}_k^1(t)] + \underset{(n \text{ even})}{\tilde{a}_q \tilde{P}_q(t)}, \quad (7)$$

where $\{\tilde{a}_k^0, \tilde{a}_k^1\}_{k=0}^{q-1}$ ($n = 2q - 1$ odd) and $\{\tilde{a}_k^0, \tilde{a}_q, \tilde{a}_k^1\}_{k=0}^{q-1}$ ($n = 2q$ even) denote the $n + 1$ coefficients of $p(t)$. The case distinction needed for the even-degree does not result in a heavy burden: the formulation is exactly that for the odd-degree plus an additional term.

Expression (7) can be rewritten in terms of the symmetric parameter $s = (1 - t)t$:

$$p(t) = \sum_{k=0}^{q-1} [(1-t)\tilde{a}_k^0 + t\tilde{a}_k^1]s^k + \underset{(n \text{ even})}{\tilde{a}_q s^q}, \quad s = (1-t)t. \quad (8)$$

The preceding summation is polynomial in monomial form in s , running over the interval $s \in [0, 1/4]$, whose coefficients are a barycentric combination of $\tilde{a}_k^0, \tilde{a}_k^1$.

2.3 A Horner's Scheme for Polynomial Evaluation

Expression (7) can also be rearranged as

$$p(t) = [(1 - t)p^0(s) + tp^1(s)] + \underset{(n \text{ even})}{\tilde{a}_q s^q}, \quad p^{0/1}(s) = \sum_{k=0}^{q-1} \tilde{a}_k^{0/1} s^k.$$

This formula contains a convex combination of two polynomials $p^0(s)$, $p^1(s)$ of degree $q - 1$ in monomial form. The evaluation of each polynomial following a Horner's scheme requires $q - 1$ additions and $q - 1$ multiplications. Adding the cost of evaluating s (1 adds, 1 mults) and the convex combination (1 adds, 2 mults), results in $2q$ adds and $2q + 1$ multiplications. For n even, the extra term $\tilde{a}_q s^q$ can be introduced into $p^0(s)$ replacing the coefficient $\tilde{a}_{q-1}^0$ by $\tilde{a}_{q-1}^0 + t\tilde{a}_q$. Hence the evaluation incurs one extra addition and multiplication. For both n even or odd, the total is $(n + 1)$ adds and $(n + 2)$ multiplications, which is almost as fast as using the standard power form.

The possibility of devising a nested multiplication algorithm admits a geometric interpretation in the space of polynomials $\tilde{P}_{i,j}(t)$. Marching to the right, upwards, or along the symmetry diagonal (Figure 1) means multiplying by t , $(1 - t)$, or s , respectively. Hence those bases arranged along these directions admit a simple Horner's algorithm. For instance, for a polynomial in the generalized Ball basis (4), a faster alternative to the recursive algorithm described by Said [1989] would be to write $p(t)$ in terms of two degree- $(q - 1)$ polynomials $p_0(t)$, $p_1(1 - t)$ in power form:

$$p(t) = (1 - t)^q p_0(t) + t^q p_1(1 - t).$$

In contrast, marching along the scaled Bernstein polynomials implies multiplying by $u = t/(1 - t)$ or $v = (1 - t)/t$. Thus a simple Horner-like scheme for polynomials in Bernstein form must resort to such rational parameter transformations [Farouki and Rajan 1988; Sederberg 1995].

2.4 Degree Operations

A polynomial $\tilde{P}_{i,j}(t)$ (1) of degree $n = i + j$ is degree-elevated by multiplying it by the unit factor $(1 - t) + t$. This operation splits the polynomial into two terms of degree $n + 1$:

$$\tilde{P}_{i,j} = \tilde{P}_{i,j+1} + \tilde{P}_{i+1,j}. \quad (9)$$

From (9) one easily gets two additional formulas that prove valuable later:

$$\tilde{P}_{i,j} = \tilde{P}_{i,j-1} - \tilde{P}_{i+1,j-1} \quad (10)$$

$$\tilde{P}_{i,j} = \tilde{P}_{i-1,j} - \tilde{P}_{i-1,j+1}. \quad (11)$$

Equation (9) can be used to transform the formulation of even-degree $n = 2q$ to that of odd-degree $2q + 1$, just by degree-elevating the symmetric polynomial (6) $\tilde{P}_q(t)$:

$$\tilde{P}_q = \tilde{P}_q^0 + \tilde{P}_q^1. \quad (12)$$

Thus the central coefficient $\tilde{a}_q$ splits as

$$\tilde{a}_q \rightarrow \begin{cases} \tilde{a}_q^0 = \tilde{a}_q \\ \tilde{a}_q^1 = \tilde{a}_q \end{cases}.$$

The actual or minimum $n_{\min}$ degree of a polynomial in the symmetric power form is simply the degree of the basis function associated with the nonzero coefficient(s) with highest subscript, except when these coefficients are the pair $\tilde{a}_k^0 = \tilde{a}_k^1$ and, consequently, the polynomial can be degree-reduced to $n_{\min} = 2k$.

In comparison, computing $n_{\min}$ of a polynomial in Bernstein form requires a more complex procedure [Farouki and Rajan 1988]. As for the generalized Ball case, pairs of central coefficients must be compared [Goodman and Said 1991a], which is less involved, but not as simple as for the symmetric power form. Observe that the generalized Ball polynomials trace out two line segments equidistant from the Bernstein and symmetric power polynomials (see Figures 1 and 2) and, in consequence, it is no surprise that they behave in an intermediate way.

2.5 Normalized Symmetric Power Polynomials

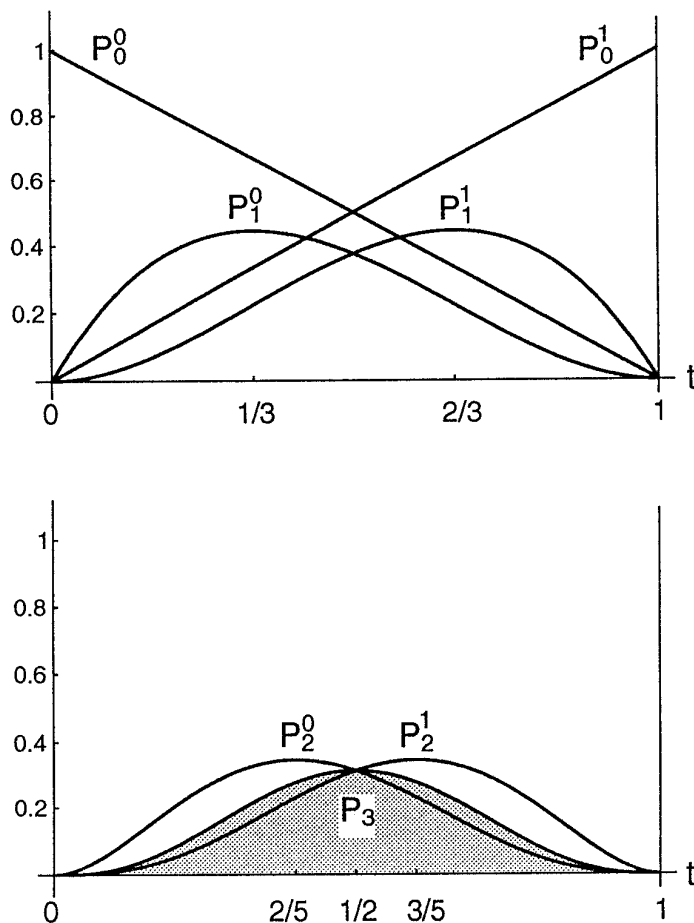
The symmetric power polynomials are, in fact, certain scaled Bernstein polynomials. Therefore it makes sense to define the normalized symmetric power polynomials that include the binomial coefficients of such Bernstein polynomials. Thus the normalized basis functions for the odd-degree $n = 2q - 1$ are:

$$\left. \begin{aligned} P_k^0(t) = B_k^n(t) &= \binom{2k+1}{k} t^k (1-t)^{k+1} \\ P_k^1(t) = B_{n-k}^n(t) &= \binom{2k+1}{k} t^{k+1} (1-t)^k \end{aligned} \right\} k = 0, \dots, q-1. \quad (13)$$

For the even-degree $n = 2q$, we add the following polynomial to the set (13),

$$P_q(t) = B_q^n(t) = \binom{n}{q} t^q (1-t)^q. \quad (14)$$

The basis functions (13) and (14) up to degree $n = 6$ are plotted in Figure 4. They are pairs of central Bernstein polynomials for successive degrees 1, 3, 5 plus the central Bernstein polynomial of degree $n = 6$.

Fig. 4. Normalized symmetric power basis functions, $n = 6$.

The binomial factors appearing in (13) and (14) seem to be the “natural” scaling factors of the basis functions, as is corroborated in Section 5. These factors relate the normalized coefficients $\{a_k^0, a_q, a_k^1\}$ of a polynomial in the normalized basis and those in expression (7):

$$\tilde{a}_k^{0/1} = \binom{2k+1}{k} a_k^{0/1}, \quad \tilde{a}_q = \binom{n}{q} a_q. \quad (15)$$

3. CONVERSIONS BETWEEN THE BERNSTEIN AND SYMMETRIC POWER FORMS

3.1 Conversion Matrices

The conversion matrices between the Bernstein and symmetric power forms take simpler expressions using the scaled bases functions that do not include binomial numbers. Given a polynomial $p(t)$, $\tilde{b}_i$ will denote its

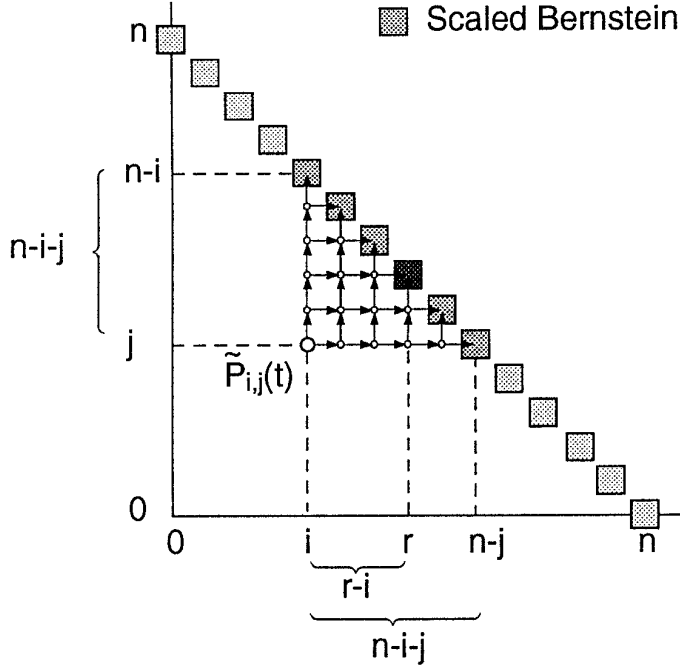


Fig. 5. Transforming a polynomial $\tilde{P}_{i,j}(t)$ to the scaled Bernstein form.

scaled Bernstein coefficients [Farouki and Rajan 1988] obtained from the standard Bernstein coefficients b_i as

$$\tilde{b}_i = \binom{n}{i} b_i. \quad (16)$$

Let $\tilde{\mathbf{b}} = (\tilde{b}_0, \dots, \tilde{b}_n)^T$ be the column vector containing the scaled Bernstein coefficients of $p(t)$, and let $\tilde{\mathbf{a}}$ be the column vector with the symmetric power coefficients:

$$n \text{ odd: } \tilde{\mathbf{a}} = (\tilde{a}_0^0, \dots, \tilde{a}_{q-1}^0, \tilde{a}_{q-1}^1, \dots, \tilde{a}_0^1)^T, \quad q = (n+1)/2$$

$$n \text{ even: } \tilde{\mathbf{a}} = (\tilde{a}_0^0, \dots, \tilde{a}_{q-1}^0, \tilde{a}_q, \tilde{a}_{q-1}^1, \dots, \tilde{a}_0^1)^T, \quad q = n/2.$$

Let $\mathbf{T}, \mathbf{T}^{-1}$ be $(n+1) \times (n+1)$ transformation matrices:

$$\tilde{\mathbf{a}} = \mathbf{T}\tilde{\mathbf{b}}, \quad \tilde{\mathbf{b}} = \mathbf{T}^{-1}\tilde{\mathbf{a}}. \quad (17)$$

As both the scaled Bernstein and symmetric power polynomials meet the symmetry condition shown in Section 2.1, the elements $T_{j,k}, T_{j,k}^{-1}$ of such matrices verify the property

$$T_{n-j,n-k} = T_{j,k}, \quad T_{n-j,n-k}^{-1} = T_{j,k}^{-1}, \quad \begin{cases} 0 \leq j \leq n \\ 0 \leq k \leq n \end{cases}. \quad (18)$$

In the following sections, explicit expressions for $T_{j,,k}$, $T_{j,k}^{-1}$ are derived.

3.2 Conversion From Symmetric Power to Scaled Bernstein Form

A given polynomial $\tilde{P}_{i,j}(t)$ can be expressed in terms of the scaled Bernstein polynomials of degree $n \geq (i + j)$ by repeated degree-elevation (9). This process has been graphically represented in Figure 5, and yields

$$\tilde{P}_{i,j}(t) = \sum_{r=i}^{n-j} \binom{n-i-j}{r-i} \tilde{B}_r^n(t). \quad (19)$$

For the symmetric power polynomials $\tilde{P}_k^0(t)$, $\tilde{P}_q(t)$ in (5) and (6) this relationship becomes

$$\tilde{P}_k^0(t) = \sum_{r=k}^{n-k-1} \binom{n-2k-1}{r-k} \tilde{B}_r^n(t)$$

$$\tilde{P}_q(t) = \tilde{B}_q^n(t) \quad (n \text{ even}).$$

The binomial numbers in the preceding summation form the k th column of $\mathbf{T}^{-1}$. Hence, taking into account the property (18), the nonzero elements $T_{j,k}^{-1}$ are:

$$T_{j,k}^{-1} = T_{n-j,n-k}^{-1} = \begin{pmatrix} n-2k-1 \\ j-k \end{pmatrix} \quad \begin{matrix} k=0, \dots, q-1 \\ j=k, \dots, n-k-1 \end{matrix}$$

$$q = \begin{cases} n/2 & (n \text{ even}) \\ (n+1)/2 & (n \text{ odd}) \end{cases}$$

$$T_{q,q}^{-1} = 1 \quad q = n/2 \quad (n \text{ even}).$$

The matrix $\mathbf{T}^{-1}$ has a unitary principal diagonal, and its structure differs slightly for degrees n even or odd. In both cases, the successive matrices for degrees n , $n+2$, $n+4$, ... are nested one into the other:

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & & & & 0 \\ 5 & \begin{bmatrix} 1 & & 0 \end{bmatrix} & & & 1 \\ 10 & 3 \begin{bmatrix} 1 & & 0 \end{bmatrix} & & & 5 \\ 10 & 3 \begin{bmatrix} 1 & & 0 \end{bmatrix} & [1] & 1 \begin{bmatrix} 3 & & 1 \end{bmatrix} & 10 \\ 5 & 1 \begin{bmatrix} 0 & & 1 \end{bmatrix} & & 3 \begin{bmatrix} 3 & & 1 \end{bmatrix} & 10 \\ 1 & \begin{bmatrix} 0 & & 1 \end{bmatrix} & & 1 & 5 \\ 0 & & & & 1 \end{bmatrix} \quad \begin{matrix} n \text{ even: } 0,2,4,6 \end{matrix}$$

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & & & & 0 \\ 6 & \begin{bmatrix} 1 & & 0 \end{bmatrix} & & & 1 \\ 15 & 4 \begin{bmatrix} 1 & & 0 \end{bmatrix} & & & 6 \\ 20 & 6 \begin{bmatrix} 1 & & 0 \end{bmatrix} & 2 \begin{bmatrix} 1 & & 0 \end{bmatrix} & & 15 \\ 15 & 4 \begin{bmatrix} 1 & & 0 \end{bmatrix} & 0 & 1 \begin{bmatrix} 2 & & 1 \end{bmatrix} & 20 \\ 6 & 1 \begin{bmatrix} 0 & & 1 \end{bmatrix} & & 4 \begin{bmatrix} 4 & & 1 \end{bmatrix} & 15 \\ 1 & \begin{bmatrix} 0 & & 1 \end{bmatrix} & & 1 & 6 \\ 0 & & & & 1 \end{bmatrix} \quad \begin{matrix} n \text{ odd: } 1,3,5,7 \end{matrix}$$

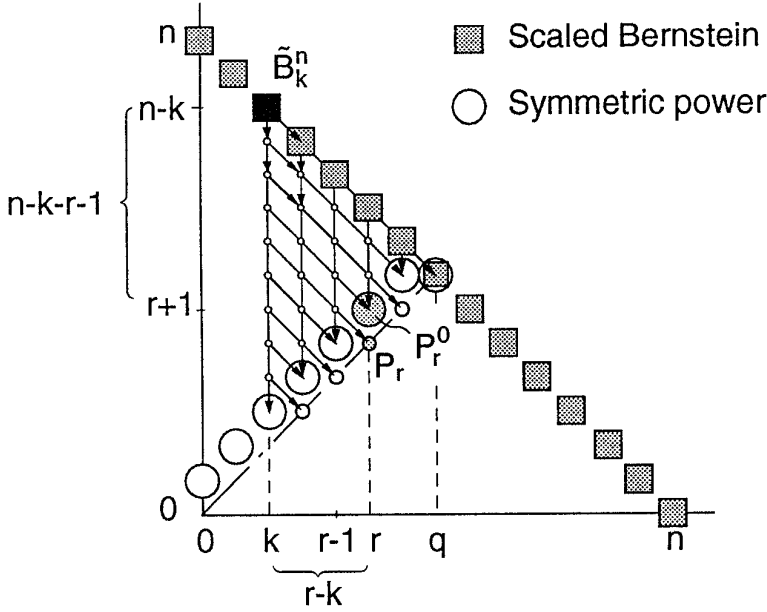


Fig. 6. Transforming the scaled Bernstein polynomial $\tilde{B}_k^n(t)$ to the symmetric power form.

3.3 Conversion From Scaled Bernstein to Symmetric Power Form

The derivation of matrix $\mathbf{T}$ is more involving than that of $\mathbf{T}^{-1}$. The k th column of $\mathbf{T}$ contains the coefficients corresponding to the expression of the k th scaled Bernstein polynomial $\tilde{B}_k^n(t)$ in the symmetric power basis. Assuming $0 \leq k < q$, these coefficients are obtained in the following steps.

- (1) Express the polynomial $\tilde{B}_k^n(t)$, marked in bold type in Figure 6, in terms of polynomials $\tilde{P}_r^0(t)$, $\tilde{P}_r(t)$ by repeated application of Equation (10):

$$\begin{aligned} \tilde{B}_k^n(t) = & \sum_{r=k}^{q-1} \binom{n-k-r-1}{r-k} (-1)^{r-k} \tilde{P}_r^0(t) + \sum_{r=k+1}^{q-1} \binom{n-k-r-1}{r-k-1} \\ & \cdot (-1)^{r-k} \tilde{P}_r(t) + (-1)^{q-k} \tilde{P}_q(t) \end{aligned} \quad (20)$$

(n even).

- (2) Degree-elevate using (12) the polynomials $\tilde{P}_r(t)$ appearing in the second summation of (20):

$$\tilde{P}_r(t) = \tilde{P}_r^0(t) + \tilde{P}_r^1(t).$$

- (3) Introduce the resulting polynomials $\tilde{P}_r^0(t)$ into the first summation of

(20) by applying the Pascal triangle rule for binomial coefficients:

$$\binom{x-1}{y} + \binom{x-1}{y-1} = \binom{x}{y}, \quad \begin{array}{l} x = n - k - r \\ y = r - k. \end{array}$$

Thus Equation (20) is transformed into

$$\begin{aligned} \tilde{B}_k^n(t) = & \sum_{r=k}^{q-1} \binom{n-k-r}{r-k} (-1)^{r-k} \tilde{P}_r^0(t) + \sum_{r=k+1}^{q-1} \binom{n-r-k-1}{r-k-1} \\ & \cdot (-1)^{r-k} \tilde{P}_r^1(t) + (-1)^{q-k} \tilde{P}_q(t) \\ & (n \text{ even}). \end{aligned}$$

For the k th column, the term multiplying the basis function $\tilde{P}_r^0(t)$ corresponds to the r th row, that multiplying $\tilde{P}_r^1(t)$ is in the $j = (n - r)$ th row, and that multiplying $\tilde{P}_q(t)$ (only appears if n is even) goes to the central q th row. Thus, taking into account the property (18), the nonzero elements $T_{j,k}$ are:

$$k = 0, \dots, q - 1$$

$$\left\{ \begin{array}{ll} T_{j,k} = T_{n-j,n-k} = (-1)^{j-k} \binom{n-j-k}{j-k}, & j = k, \dots, q - 1 \\ T_{q,k} = T_{q,n-k} = (-1)^{q-k}, \quad T_{q,q} = 1 & (n \text{ even}) \\ T_{n-j,k} = T_{j,n-k} = (-1)^{j-k} \binom{n-j-k-1}{j-k-1}, & j = k + 1, \dots, q - 1. \end{array} \right.$$

The matrix $\mathbf{T}$ shows a structure similar to that of $\mathbf{T}^{-1}$, but now the nonzero elements are alternating in sign.

3.4 Conversion Matrices Between the Bernstein and Normalized Symmetric Power Form

Consider now the standard Bernstein basis and the normalized symmetric power basis, which include the binomial factors. Let $\mathbf{b}$, $\mathbf{a}$ be column vectors containing the respective coefficients in these bases of a polynomial, and let $\mathbf{M}$, $\mathbf{M}^{-1}$ be the transformation matrices:

$$\mathbf{a} = \mathbf{M}\mathbf{b}, \quad \mathbf{b} = \mathbf{M}^{-1}\mathbf{a}. \quad (21)$$

These matrices enjoy a symmetry property similar to (18). Thus the nonzero elements $\mathbf{M}_{j,k}$ of $\mathbf{M}$ are generated from those of $\mathbf{T}$ just by including

the binomial factors appearing in (15) and (16):

$$k = 0, \dots, q - 1$$

$$\left\{ \begin{array}{l} M_{j,k} = M_{n-j,n-k} = (-1)^{j-k} \binom{n-j-k}{j-k} \binom{n}{k} \binom{2j+1}{j}^{-1} \\ \qquad \qquad \qquad j = k, \dots, q - 1 \\ \\ M_{q,k} = M_{q,n-k} = (-1)^{q-k} \binom{n}{k} \binom{n}{q}^{-1} \quad M_{q,q} = 1 \\ \qquad \qquad \qquad (n \text{ even}) \\ \\ M_{n-j,k} = M_{j,n-k} = (-1)^{j-k} \binom{n-j-k-1}{j-k-1} \binom{n}{k} \binom{2j+1}{j}^{-1} \\ \qquad \qquad \qquad j = k + 1, \dots, q - 1. \end{array} \right.$$

The nonzero elements of $\mathbf{M}^{-1}$ are derived in a similar way:

$$\begin{aligned} M_{j,k}^{-1} &= M_{n-j,n-k}^{-1} = \binom{n-2k-1}{j-k} \binom{2k+1}{k} \binom{n}{j}^{-1} \quad \left\{ \begin{array}{l} k = 0, \dots, q - 1 \\ j = k, \dots, n - k - 1 \end{array} \right. \\ M_{q,q}^{-1} &= 1 \quad (n \text{ even}). \end{aligned} \tag{22}$$

3.5 Numerical Condition of the Problem

The numerical condition of a problem is a measure of the intrinsic sensitivity of its solution to perturbations in its input parameters. The numerical condition is characterized by means of scalar values called condition numbers, which indicate the maximum possible error in the solution under random input perturbations. These perturbations will certainly occur if a floating-point representation is used. Needless to say, problems with large condition numbers, so-called ill-conditioned problems, must be avoided.

An important issue is the numerical condition of the transformation of polynomial coefficients between two different bases. It is well known that this problem is ill-conditioned for the transformation between the power and Bernstein forms [Daniel and Daubisse 1989; Farouki 1991], because the condition of the problem deteriorates rapidly as the degree n increases. In other words, small errors in the coefficients in the original basis may be enormously amplified by the transformation. Therefore we must check that this does not occur for transformation between the symmetric power and Bernstein forms.

Suppose that a given polynomial of degree n has coefficients $\mathbf{c}_\phi = (c_0, \dots, c_n)^T$ in a certain basis $\{\phi_k(t)\}$, and that these coefficients are mapped into the coefficients $\mathbf{c}_\psi = \mathbf{M}\mathbf{c}_\phi$ in another basis $\{\psi_k(t)\}$ by the transformation matrix $\mathbf{M}$. If the coefficients $\mathbf{c}_\phi$ suffer a perturbation $\delta\mathbf{c}_\phi = (\delta c_0, \dots, \delta c_n)^T$, the corresponding perturbation induced in the trans-

formed coefficients will be $\delta \mathbf{c}_\psi = \mathbf{M} \delta \mathbf{c}_\phi$. These perturbations can be characterized by relative scalar measurements:

$$\varepsilon_\phi = 10^{-\lambda_\phi} = \frac{\|\delta \mathbf{c}_\phi\|_p}{\|\mathbf{c}_\phi\|_p}, \quad \varepsilon_\psi = 10^{-\lambda_\psi} = \frac{\|\delta \mathbf{c}_\psi\|_p}{\|\mathbf{c}_\psi\|_p},$$

where $\|\mathbf{c}\|_p$ denotes the p -norm of a vector,

$$\|\mathbf{c}\|_p = \left[\sum_{k=0}^n |c_k|^p \right]^{1/p}. \quad (23)$$

The norms most frequently employed are $p = 1$, $p = 2$ (Euclidean norm) and $p = \infty$ (maximum norm).

As shown by Farouki [1991], the induced error ε_ψ is bounded in terms of ε_ϕ by

$$\varepsilon_\psi \leq \kappa_p(\mathbf{M}) \varepsilon_\phi, \quad \kappa_p(\mathbf{M}) = \|\mathbf{M}\|_p \|\mathbf{M}^{-1}\|_p \quad (24)$$

or, in logarithmic form:

$$\lambda_\psi \leq \lambda_\phi - \lambda_p, \quad \lambda_p = \log_{10}[\kappa_p(\mathbf{M})], \quad (25)$$

where $\kappa_p(\mathbf{M})$ denotes the condition number of the matrix $\mathbf{M}$ in the p -norm (23), and $\|\mathbf{M}\|$ is the matrix norm subordinate to the vector norm. For $p = 1$, $p = \infty$, which are the most amenable cases for computation,

$$\|\mathbf{M}\|_1 = \max_{0 \leq k \leq n} \sum_{j=0}^n |M_{j,k}|, \quad \|\mathbf{M}\|_\infty = \max_{0 \leq j \leq n} \sum_{k=0}^n |M_{j,k}|.$$

3.6 Values of the Condition Numbers. Practical Consequences

Figure 7 plots the logarithms λ_1 , λ_∞ (25) of the condition numbers $\kappa_1(\mathbf{M})$, $\kappa_\infty(\mathbf{M})$ (24) as a function of the degree n for the transformation between the Bernstein and the symmetric power form. It is observed that $\kappa_1 \approx \kappa_\infty$. This figure also shows the condition numbers corresponding to the transformation between the Bernstein and monomial forms, for which $\kappa_1 = \kappa_\infty$ [Farouki 1991]. The transformation involving the symmetric form is much better conditioned, with λ_1 or λ_∞ roughly one-third of that for the monomial basis.

Rather than the condition numbers, a more practical figure is the number n_∞ of significant digits of the most affected transformed coefficient $\mathbf{c}_\psi$. As demonstrated by Daniel and Daubisse [1989], in the worst case this number takes the value:

$$n_\infty \approx \max(0, \text{round}(\lambda_\phi - \lambda_\infty)), \quad (26)$$

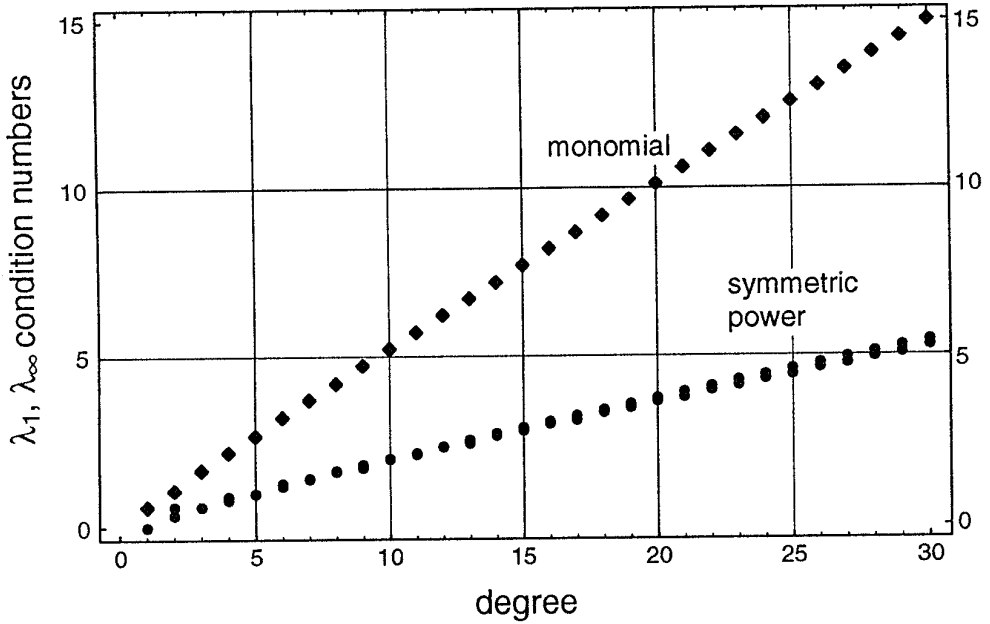


Fig. 7. Logarithms $\lambda_1, \lambda_\infty$ of the condition numbers κ_p ($p = 1$ and $p = \infty$).

where the function $\text{round}(x)$ returns the integer closest to x . In other words, λ_∞ indicates roughly the worst-case loss of significant digits due to the basis conversion.

As an illustrative example, we consider perturbations of the order of the last bit of the mantissa in the floating-point representation of $\mathbf{c}_\phi$, that is, rounding errors. Assuming the customary mantissas of 24 and 56 bits for single and double precision, respectively, the corresponding perturbations are $\varepsilon_\phi = 2^{-24}$ ($\lambda_\phi = 7.22$) and $\varepsilon_\phi = 2^{-56}$ ($\lambda_\phi = 18.66$). Figure 8 plots the number of significant digits given by (26) in this example as a function of the degree n . It must be stressed that a magnitude of the number of significant digits is obtained, not precise values. The results are much better than those corresponding to the transformation between the Bernstein and monomial forms [Daniel and Daubisse 1989], for which the number of significant digits that are lost is three times larger. It can be concluded that a reasonably accurate conversion between the Bernstein and symmetric power form is feasible.

4. CONDITION NUMBERS FOR EVALUATION AND ROOT FINDING

4.1 Condition Numbers for the Evaluation of a Polynomial

A polynomial basis for CAGD purposes must not be prone to magnify small errors in its coefficients when a curve or surface is evaluated. This sensitivity is characterized by means of the condition numbers for evaluation.

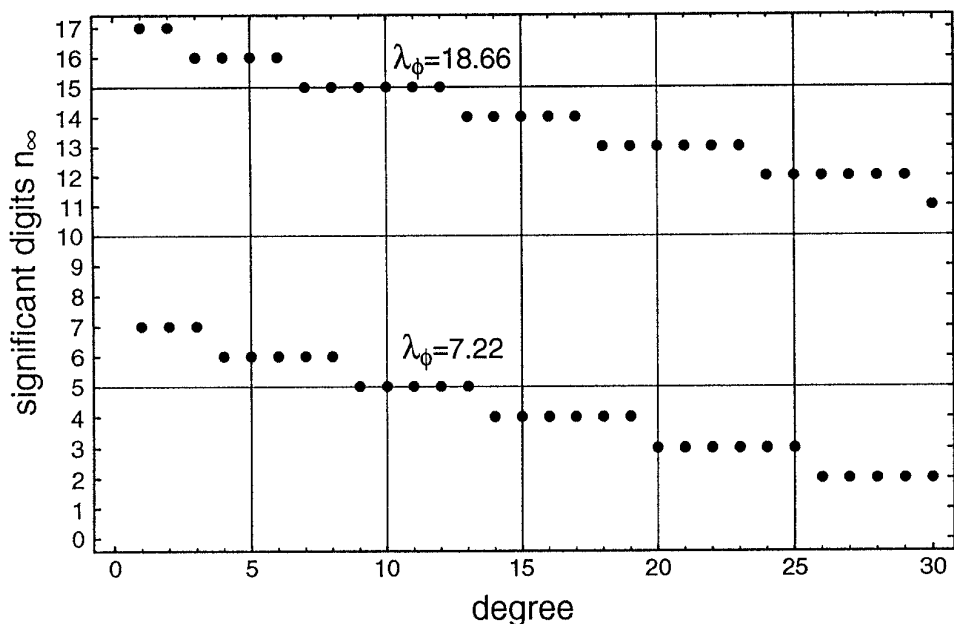


Fig. 8. Minimum number of significant digits for a transformed polynomial.

Consider a certain degree- n polynomial basis $\{\phi_k(t)\}$ over $t \in [0, 1]$, and suppose that a given polynomial $p(t)$ has coefficients c_k in this basis:

$$p(t) = \sum_{k=0}^n c_k \phi_k(t).$$

If all the coefficients suffer a perturbation of fixed relative magnitude $\varepsilon = |\delta c_k / c_k|$, then the magnitude $|\delta p|$ of the change induced in $p(t)$ can be bounded as

$$|\delta p| \leq \varepsilon S_\phi(t), \quad S_\phi(t) = \sum_{k=0}^n |c_k \phi_k(t)|. \quad (27)$$

The sum $S_\phi(t)$ is called the condition number for the evaluation of the polynomial $p(t)$ at value t [Farouki and Rajan 1987], and indicates the worst-case amplification factor of the perturbation. Note that this number does not depend on the evaluation algorithm of $p(t)$, as it is assumed to operate in infinite-precision arithmetic. The sum $S_\phi(t)$ depends on the chosen polynomial basis, but it is independent of the particular scaling adopted for the individual basis functions $\{\phi_k(t)\}$. In other words, in a new basis

$$\psi_k(t) = s_k \phi_k(t), \quad k = 0, \dots, n,$$

where $\{s_k\}$ are arbitrary nonzero scale factors, it is verified that $S_\psi(t) = S_\phi(t)$. Therefore the scaled Bernstein (3) and the Bernstein bases are completely equivalent as to condition numbers. The same holds true for the symmetric power (5) and (6) and normalized (13) and (14) forms.

4.2 Condition Number for Polynomial Root

In many geometry processing algorithms [Barnhill 1992] the real roots of polynomials must be computed. Hence it is desirable to use a basis such that the sensitivity of the roots to perturbations in the coefficients is low. This numerical property is characterized by means of the condition number for polynomial roots. Assume that a polynomial $p(t)$ has a simple root $t_0 \in [0, 1]$, which implies a nonnull derivative $p'(t_0)$. If all the coefficients suffer a fixed perturbation $\varepsilon = |\delta c_k/c_k|$, the displacement $|\delta t|$ a simple root suffers in the limit $\varepsilon \rightarrow 0$ can be bounded as [Farouki and Rajan 1987]

$$|\delta t| \leq \varepsilon C(t_0), \quad C(t_0) = |p'(t_0)|^{-1} S_\phi(t_0), \quad (28)$$

where $C(t_0)$ is called the condition number of the root t_0 . There exists a similar definition for multiple roots.

4.3 Comparing the Condition Numbers for Different Polynomial Bases

This section presents a comparative study of the condition numbers for different bases (Bernstein, generalized Ball, monomial, and symmetric power) constructed by using polynomials (1) $P_{i,j}(t)$. It must be stressed that the derivative appearing in the definition (28) does not depend on the polynomial basis that has been used. Thus, comparing $C(t_0)$ for different bases reduces to a comparison among values $S_\phi(t_0)$.

First the condition number for a polynomial with a root at $t_0 = 0$ is examined. The only polynomials $P_{i,j}(t)$ not vanishing at $t_0 = 0$ are those with $i = 0$ and the four considered bases contain just one of these polynomials. Hence for such bases the corresponding coefficient must be $c_k = 0$, all the terms in the sum (27) must be zero, $S_\phi(0) = 0$, and $C(0) = 0$. Stated informally, these bases enjoy good numerical properties near the origin. In general, the condition that a basis of polynomials $P_{i,j}(t)$ be well-conditioned at $t_0 = 0$ is that only one basis function lies on the vertical axis $i = 0$. Analogously, for a root at $t_0 = 1$ the requirement is that only one basis function lies on the horizontal axis $j = 0$, so that $S_\phi(1) = 0$ and $C(1) = 0$. This condition is met by the Bernstein, generalized Ball, and symmetric power basis, but not by the monomial basis. Thus the symmetric power form possesses excellent numerical properties at the endpoints of $[0, 1]$, and the discontinuities in the piecewise case (see the Introduction) are completely avoided.

As for the interval $t \in (0, 1)$, a theorem demonstrated in Farouki and Rajan [1987] is very valuable. It states that, for any two polynomial bases nonnegative over a given interval, if one can be expressed as a nonnegative linear combination of the other, then the latter always provides better

root-condition on that interval than the former. Expression (19) indicates that any polynomial $P_{i,j}(t)$ of degree $\leq n$ can be written as a nonnegative combination of Bernstein polynomials of degree n . It is concluded that the Bernstein basis is the best root-conditioned basis that can be constructed with polynomials $P_{i,j}(t)$. Similarly, the generalized Ball basis is better conditioned than the symmetric power basis for $t \in (0, 1)$, as any symmetric power polynomial can be expanded by repeated application of (9) into scaled Ball polynomials. Intuitively speaking, those bases composed of polynomials $\tilde{P}_{i,j}(t)$ far from the origin (high values $i + j$) are better root-conditioned than those close to the origin. Thus the convenience of a set of basis functions with increasing degrees comes at the expense of higher condition numbers.

Nonetheless, there are situations in which the symmetric power form provides better actual accuracy than the Bernstein form. For instance, consider a linear polynomial that, for some reason, must be expressed as a polynomial of higher degree. Using the Bézier form in a floating-point environment this degree elevation will degrade accuracy because the new coefficients are obtained through floating-point operations, and because the error induced by the floating-point operations in the evaluation algorithm will certainly grow as a function of the degree. Note that these facts are not reflected by the condition numbers. In contrast, degree elevation in the symmetric power form reduces to adding a lot of zero coefficients, which does not affect the accuracy of computations.

A transformation between the monomial and symmetric power forms would make use of the relationships (10) and (11), which would not lead to nonnegative combinations. Hence the aforementioned theorem cannot be applied. Nevertheless, from (8) the following inequality holds over $t \in (0, 1)$ for the condition number $S_s(t)$ (27) of a degree- n polynomial in symmetric power form.

$$S_s(t) \leq \sum_{k=0}^{q-1} c_k s^k + |\tilde{a}_q| s^q, \quad c_k = \max(|\tilde{a}_k^0|, |\tilde{a}_k^1|). \quad (n \text{ even})$$

Thus $S_s(t)$ is lower than the condition number of a certain polynomial in monomial form. This polynomial has a domain restricted to the interval $s \in (0, 1/4)$, which is the stable area near the origin. Furthermore it has a degree reduced to one-half of n , thereby enjoying additional stability. In consequence, for an arbitrary polynomial, it is reasonable to expect that the symmetric power form enjoys much better numerical condition than the power form, as corroborated by the following examples.

4.4 Some Illustrative Examples

Example 1. Figure 9 compares the influence of perturbations in the polynomial coefficients for the power and symmetric power bases, following

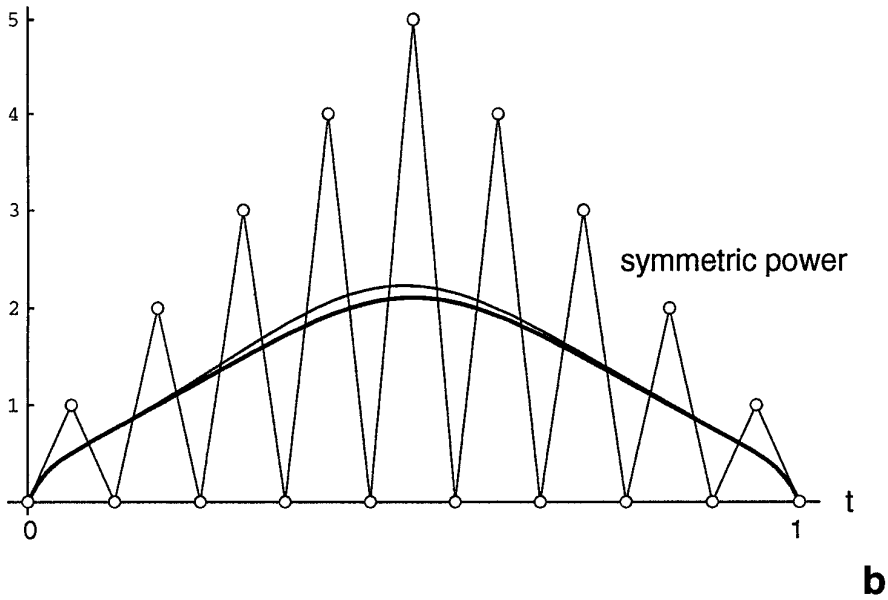
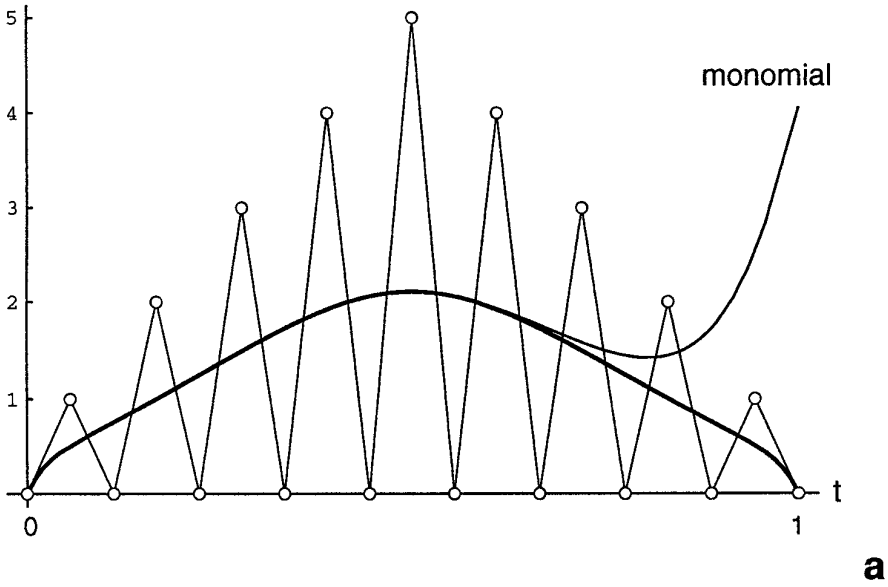


Fig. 9. Influence of perturbations in the coefficients of a polynomial (degree $n = 18$): (a) $\varepsilon = 10^{-8}$; (b) $\varepsilon = 10^{-3}$.

an example in Farin [1993]. The polynomial (degree $n = 18$) has been defined in the Bézier form, with coefficients

$$\mathbf{b} = \{0, 1, 0, 2, 0, 3, 0, 4, 0, 5, 0, 4, 0, 3, 0, 2, 0, 1, 0\}.$$

Next the polynomial has been transformed to both the power and symmetric power form using the corresponding transformation matrices, and all coefficients in these forms have been modified, with perturbations of random signs and fixed relative magnitude ε . Finally, the perturbed polynomials have been displayed. The transformations from the original Bézier polynomial and the polynomial evaluation have been performed in exact rational arithmetic in the Mathematica computer algebra system [Wolfram 1993]. Setting $\varepsilon = 0$ yields identical plots for all forms, so that the displayed perturbations stem exclusively from the perturbation in the coefficients.

For $\varepsilon = 10^{-8}$ (Figure 9a) the symmetric power form coincides with the original Bézier polynomial, whereas the monomial form is not reliable near $t = 1$. For $\varepsilon = 10^{-3}$ (Figure 9b) the power form (not plotted) goes out of scale, and the symmetric power form shows some visual signs of perturbations in the central part of the interval $[0, 1]$, as expected.

Example 2. This example compares the root condition numbers for different bases, following an example in Farouki and Rajan [1987]. The polynomial $p(t)$ to test has degree $n = 20$ and roots $t_0 = \{k/n\}_{k=1}^n$ uniformly distributed on $[0, 1]$:

$$p(t) = \prod_{k=1}^{20} (t - k/20).$$

This product was expanded in the power basis, and then transformed to both the Bernstein and symmetrical power forms, using exact rational arithmetic. The corresponding root condition numbers (28) were obtained in rational form, and finally a floating-point approximation was computed by means of a single division. Figure 10 plots the results for the three bases. It is observed that the symmetric power basis displays condition numbers closer to the Bernstein form, and both provide a substantial improvement with respect to the monomial form.

Next all the coefficients in both the power and Bernstein forms were submitted to the same relative perturbation $\varepsilon = 10^{-12}$, to analyze what happened to the roots of the perturbed polynomials. Numerical approximations of these roots are listed in Table I. Those digits reflecting the displacement of a root are stroked through, so that the magnitude of such a displacement becomes apparent. In the monomial case the roots $k = 9, \dots, 20$ suffer massive displacements and now are complex, whereas in the Bernstein and symmetric power forms they remain real and close to their original locations. As predicted by the root-condition numbers, the Bernstein form yields the best results.

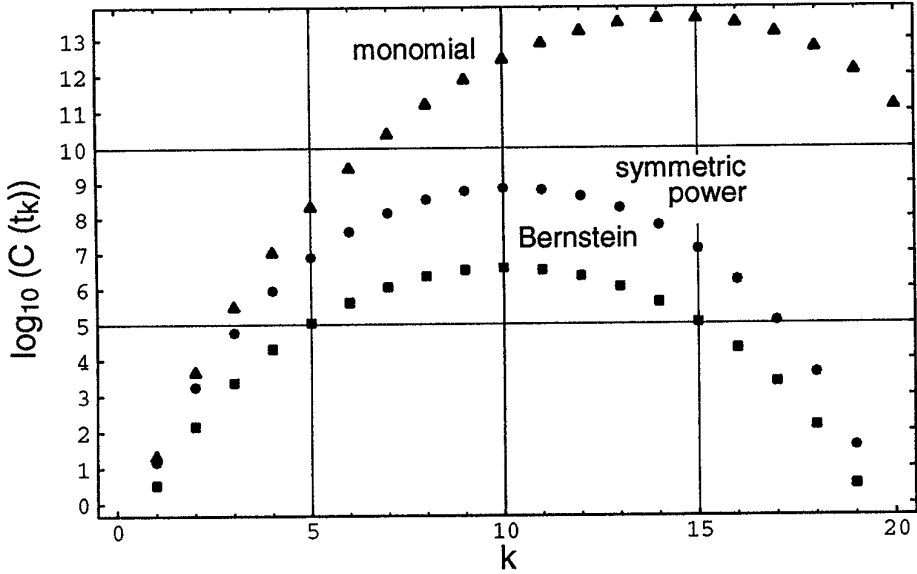


Fig. 10. Root conditions numbers for Example 2.

5. GEOMETRIC PROPERTIES

5.1 Correspondence to Forrest's Degree-Reduction Scheme

In this section, it will be checked that truncating the higher terms in the symmetric power form of a polynomial $p(t)$ is equivalent to invoking successive degree reductions following Forrest's [1972] algorithm. This process yields quite reasonable approximations of $p(t)$: successive Hermite interpolants of its derivatives at the endpoints $t = 0, t = 1$.

Assume that $p(t)$ has coefficients $\{a_k^0, a_k^1\}$ (odd-degree $n = 2q - 1$) or $\{a_k^0, a_q, a_k^1\}$ (even-degree $n = 2q$) in the normalized symmetric power basis (13) and (14):

$$p(t) = \sum_{k=0}^{q-1} [a_k^0 P_k^0(t) + a_k^1 P_k^1(t)] + \underbrace{a_q P_q(t)}_{(n \text{ even})}. \quad (29)$$

The Hermite interpolant $h^k(t)$ of order $k < q$ is the polynomial of degree $(2k + 1)$ that reproduces the derivatives $\{p^{(r)}(0), p^{(r)}(1)\}_{r=0}^k$. As those bases of subscript $> k$ do not contribute to the prescribed derivatives, $h^k(t)$ is obtained simply by truncation of (29):

$$h^k(t) = \sum_{i=0}^k [a_i^0 P_i^0(t) + a_i^1 P_i^1(t)].$$

Table I. Roots of Perturbed Polynomials

k	<u>power</u>	<u>symmetric power</u>	<u>Bernstein</u>
1	0.050000000001	0.050000000000	0.050000000000
2	0.099999999999	0.099999999965	0.099999999996
3	0.14999999193	0.15000000023	0.15000000001
4	0.2000037557	0.2000000706	0.2000000030
5	0.2499331344	0.2499988480	0.2499999608
6	0.3005012612	0.3000075314	0.3000001777
7	0.3402602909	0.3499724240	0.3499995440
8	0.3935065205	0.4000630032	0.4000006916
9	0.4440060001 ±	0.4499065067	0.4499994606
10	0.0470560970 I	0.5000899607	0.4999999353
11	0.5370290034 ±	0.5499471937	0.5500006073
12	0.4015035329 I	0.6000130900	0.5999991300
13	0.6550180050 ±	0.6500035039	0.6500006012
14	0.1376945309 I	0.6999956636	0.6999997304
15	0.7097647912 ±	0.7500015072	0.7500000704
16	0.1427006902 I	0.7999997070	0.7999999000
17	0.9103467954 ±	0.8500000276	0.8500000010
18	0.1056364304 I	0.8999999909	0.9000000000
19	1.0000920030 ±	0.9500000000	0.9500000000
20	0.0351927934 I	1.0000000000	1.0000000000

On the other hand, Eck [1993] noted that the Hermite interpolant results from the application of Forrest's degree-reduction scheme to polynomials in Bézier form. It is concluded that, for n odd, the term

$$p(t) - h^{q-1}(t) = a_q P_q(t) \quad (30)$$

must be that neglected in the degree-reduction $n \rightarrow (n - 1)$, from $p(t)$ to the Hermite interpolant of order $q - 1$. The successive k th terms in the summation (29) are those neglected in successive degree reductions $(2k + 1) \rightarrow (2k - 1)$, from the Hermite interpolant of order k to that of order $k - 1$:

$$h^k(t) - h^{k-1}(t) = a_k^0 P_k^0(t) + a_k^1 P_k^1(t), \quad 1 \leq k < q. \quad (31)$$

It is easily checked that an intermediate degree-reduction $(2k + 1) \rightarrow 2k$ following Forrest's algorithm generates the polynomial:

$$h^{k-1}(t) + a_k P_k(t), \quad \tilde{a}_k = 1/2 (\tilde{a}_k^0 + \tilde{a}_k^1).$$

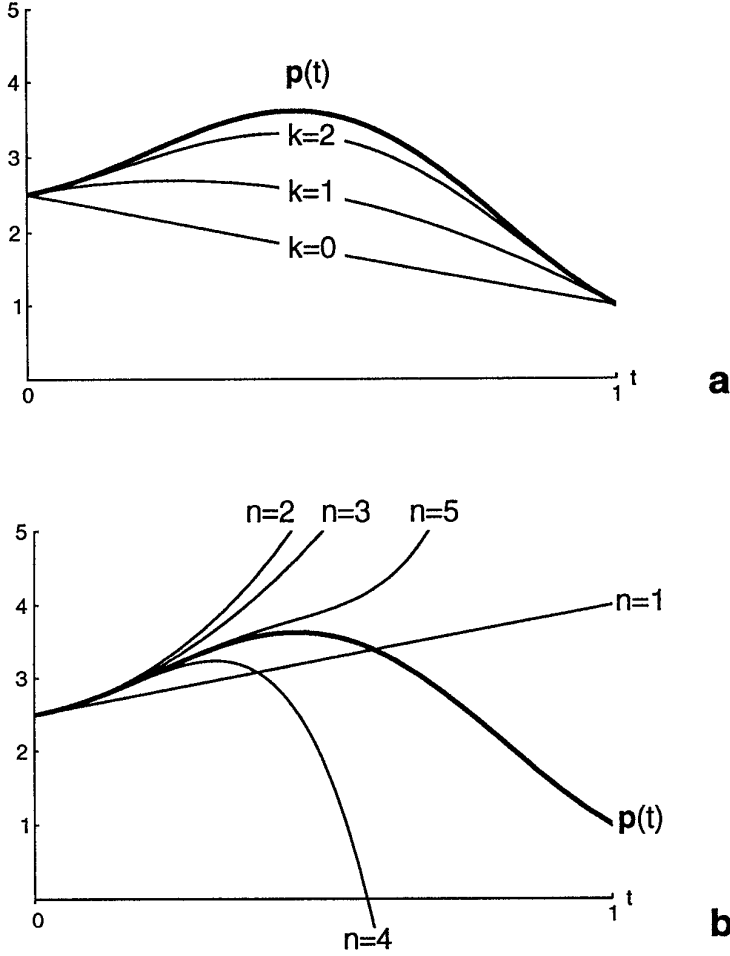


Fig. 11. Successive truncations of (a) symmetric power and (b) monomial forms. Polynomial of degree $n = 6$ and coefficients $\mathbf{a} = \{5/2, 1, 1, 1, 3/2, 1, 1\}$.

The error analysis for the degree-reduction process is therefore much simpler in the symmetric power basis than in the Bernstein [Piegl and Tiller 1995], or in the Ball basis [Goodman and Said 1991b].

Figure 11a plots the Hermite interpolants of order $k = 0, 1, 2$ for a polynomial $p(t)$ of degree $n = 6$. In contrast, Figure 11b illustrates that successive truncations of the monomial form yield very poor approximations. For a piecewise polynomial, the resulting piecewise approximations would not be even continuous, whereas successive Hermite interpolants of order k would obviously preserve C^k continuity at the joints. Nevertheless, it is worth mentioning that Forrest's degree reduction is not optimal, in the sense of minimizing the maximum error between the original polynomial and its degree-reduced approximation. To achieve this goal, Chebyshev

polynomials must be employed [Watkins and Worsey 1988; Lachance 1988; Eck 1993].

5.2 Interpreting the Normalized Coefficients

For n even, the geometric interpretation of the central coefficient a_q is obtained after rewriting (30) using Bernstein polynomials:

$$p(t) - h^{q-1}(t) = a_q B_q^n(t). \quad (32)$$

If $\{b_i\}_{i=0}^n$ denotes the Bézier coefficients of $p(t)$ and $\{\hat{h}_i^{q-1}\}_{i=0}^n$ those of the Hermite interpolant $h^{q-1}(t)$ after it is degree-elevated $(n-1) \rightarrow n$, it follows from (32) that

$$b_i = \hat{h}_i^{q-1} \begin{cases} i = 0, \dots, q-1 \\ i = q+1, \dots, n \end{cases}, \quad a_q = b_q - \hat{h}_q^{q-1}. \quad (33)$$

Figure 12a illustrates the geometry behind Equation (33). If we consider the n th degree nonparametric Bézier polygons of both $p(t)$ and its degree-elevated Hermite interpolant $\hat{h}^{q-1}(t)$ of order $q-1$, all their control points coincide, except the central one. The normalized coefficient a_q , represented by an arrow in the figure, indicates the difference between the central Bézier ordinate b_q of $p(t)$ and that $\hat{h}_q^{q-1}$ of $\hat{h}^{q-1}(t)$.

Analogously, if $\{h_i^k\}_{i=0}^{2k+1}$ denotes the Bézier coefficients of $h^k(t)$ and $\{\hat{h}_i^{k-1}\}_{i=0}^{2k+1}$ those of $h^{k-1}(t)$ after it is degree-elevated $(2k-1) \rightarrow (2k+1)$, from (31) it follows that

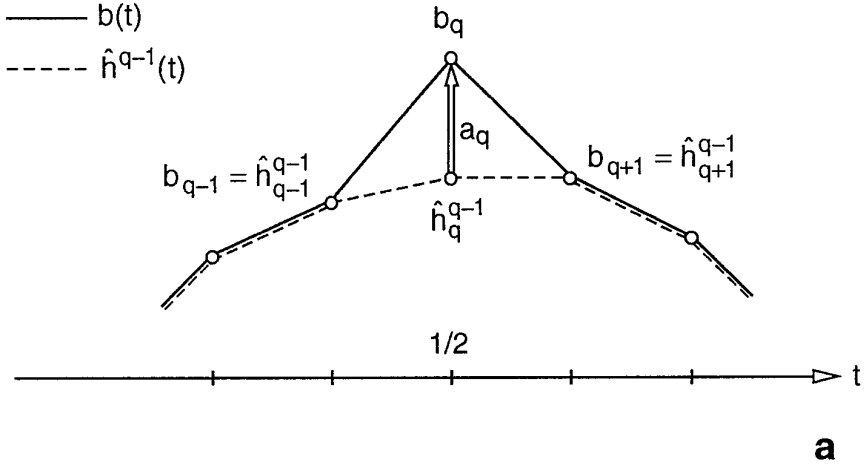
$$h_i^k = \hat{h}_i^{k-1} \begin{cases} i = 0, \dots, k-1 \\ i = k+1, \dots, 2k+1 \end{cases}, \quad \begin{aligned} a_k^0 &= h_k^k - \hat{h}_k^{k-1} \\ a_k^1 &= h_{k+1}^k - \hat{h}_{k+1}^{k-1} \end{aligned}, \quad k = 1, \dots, q-1. \quad (34)$$

Hence, the normalized coefficients $a_k^{0/1}$ can be interpreted as the difference between the central Bézier coefficients of the Hermite interpolant of order k and those of the interpolant of order $k-1$, when the latter is expressed in Bézier form of degree $(2k+1)$ (see Figure 12b). Finally, $a_0^{0/1}$ are the values of $p(t)$ at the domain endpoints:

$$a_0^0 = h_0^0 = p(0), \quad a_0^1 = h_1^0 = p(1). \quad (35)$$

Figure 13 shows how the successive normalized coefficients corresponding to the 6th degree polynomial $p(t)$ of Figure 11 are generated by applying this geometric interpretation. Observe that, roughly speaking, these normalized coefficients express the successive “degree components” of a polynomial over the internal unit, in a similar way to a periodic function being decomposed through frequency analysis into its sinusoidal components.

Bézier polygons



Bézier polygons

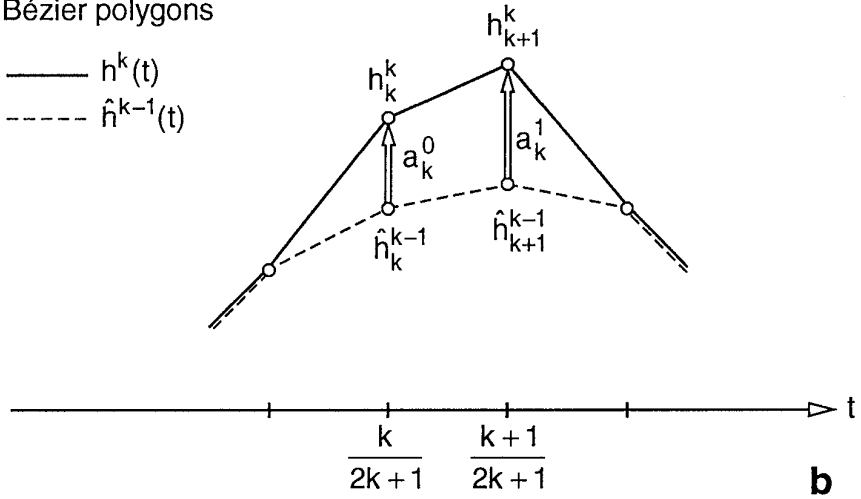


Fig. 12. Geometric interpretation of normalized coefficients (a) a_q and (b) $a_k^{0/1}$.

5.3 Using the Normalized Coefficients as Shape Handles for Curves

A parametric polynomial curve $\mathbf{p}(t)$ in the symmetric power basis takes the form

$$\mathbf{p}(t) = \sum_{k=0}^{q-1} [\mathbf{a}_k^0 P_k^0(t) + \mathbf{a}_k^1 P_k^1(t)] + \mathbf{a}_q P_q(t), \quad (n \text{ even})$$

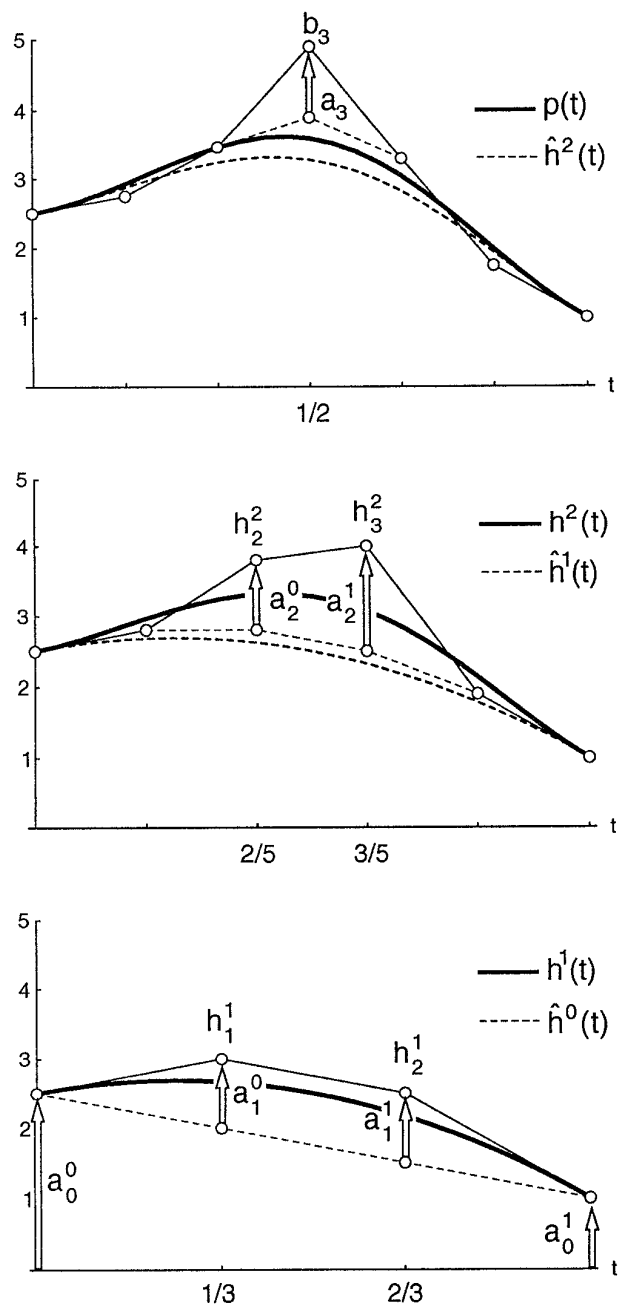


Fig. 13. Obtaining the normalized coefficients $\mathbf{a} = \{5/2, 1, 1, 1, 3/2, 1, 1\}$ for the polynomial of Figure 11.

where now all coefficients are written using boldface letters to denote that they have several components. According to (33) through (35), these coefficients can be interpreted as follows.

- $\mathbf{a}_0^0, \mathbf{a}_0^1$ are the endpoints of the curve at $t = 0, t = 1$, respectively.
- The remaining coefficients $\{\mathbf{a}_k^0, \mathbf{a}_q, \mathbf{a}_k^1\}_{k=1}^{q-1}$ are vectors, that is, differences between the Bézier control points of two curves.

The result of applying an affine transformation to $\mathbf{p}(t)$ comes as no surprise. Such a transformation Φ takes a point $\mathbf{p}$ and performs the mapping

$$\Phi(\mathbf{p}) = \mathbf{A}\mathbf{p} + \mathbf{v},$$

where $\mathbf{A}$ is a 3×3 matrix and $\mathbf{v}$ is a fixed vector. As the pair $P_0^{0/1}(t)$ form a partition of unity, the transformed curved $\Phi(p(t))$ has endpoints $\Phi(\mathbf{a}_0^{0/1})$ and vector coefficients $\{\mathbf{A}\mathbf{a}_k^0, \mathbf{A}\mathbf{a}_q, \mathbf{A}\mathbf{a}_k^1\}_{k=1}^{q-1}$. As a particular case, to translate $\mathbf{p}(t)$ along $\mathbf{v}$, just add $\mathbf{v}$ to both $\mathbf{a}_0^0, \mathbf{a}_0^1$.

As shown in (13) and (14) the normalized basis functions are certain Bernstein polynomials. It is well known that the Bernstein polynomial $B_i^n(t)$ attains maximum at $t = i / n$. Hence, the basis functions $P_k^0(t)$, $P_q(t)$, $P_k^1(t)$ attain maxima, respectively, at the following parameter values.

$$t_k^0 = k/(2k + 1), \quad t_q = 1/2, \quad t_k^1 = 1 - t_k^0. \quad (36)$$

The basis functions superscripted 0 attain maxima on the left half $t \in [0, 1/2)$, and those with superscript 1 on the right half $t \in (1/2, 1]$. As k increases, values $t_k^{0/1}$ tend to jam around $t = 1/2$. Thus, if $\mathbf{a}_k^0, \mathbf{a}_k^1$ or $\mathbf{a}_q$ is incremented by $\mathbf{v}$, this has a very predictable effect on the curve: it is pulled along the direction of $\mathbf{v}$ mostly in the neighborhood of the corresponding parameter value (36). This is illustrated in Figure 14 for the nonparametric curve previously used in Figures 11 and 13. A filled plot represents the basis function associated with each coefficient. Coefficients $\mathbf{a}_k^0$ exert more influence on one half of the curve, whereas coefficients $\mathbf{a}_k^1$ would exert more influence on the other half.

5.4 Associated Control Polygon

Rather than directly using the control vectors $\mathbf{a}_k^{0/1}, \mathbf{a}_q$ as incremental push-pull handles, it may be desirable to utilize control points associated with them, because control points allow a more intuitive design with a locator device (such as a mouse). For this purpose, one can take pairs $\mathbf{h}_k^k, \mathbf{h}_{k+1}^k$ (34) of inner Bézier points corresponding to successive Hermite interpolants, plus (case n even) the central Bézier point $\mathbf{b}_q$ of the curve $\mathbf{p}(t)$. These are the points indicated by the arrows in Figures 12 and 13. Rewriting

$$\mathbf{b}_k^0 = \mathbf{h}_k^k, \quad \mathbf{b}_k^1 = \mathbf{h}_{k+1}^k,$$

such points define a certain polygon P_k associated with the interpolant

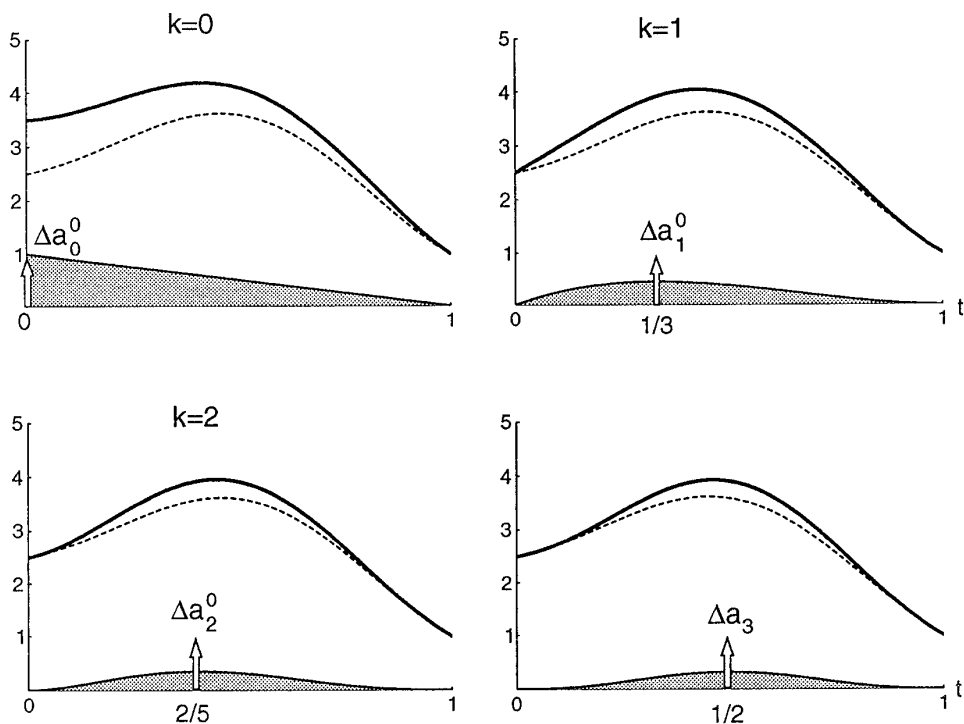


Fig. 14. Effect of modifying the normalized coefficients. Nonparametric curve (degree $n = 6$).

$\mathbf{h}^k(t)$:

$$P_k = \{\mathbf{b}_0^0, \mathbf{b}_1^0, \dots, \mathbf{b}_k^0, \mathbf{b}_k^1, \dots, \mathbf{b}_1^1, \mathbf{b}_0^1\} \quad (37)$$

and a polygon P associated with $\mathbf{p}(t)$:

$$\begin{aligned} n = 2q - 1: \quad P &= P_{q-1} = \{\mathbf{b}_0^0, \dots, \mathbf{b}_{q-1}^0, \mathbf{b}_{q-1}^1, \dots, \mathbf{b}_0^1\} \\ n = 2q: \quad P &= \{\mathbf{b}_0^0, \dots, \mathbf{b}_{q-1}^0, \mathbf{b}_q, \mathbf{b}_{q-1}^1, \dots, \mathbf{b}_0^1\}. \end{aligned} \quad (38)$$

Figure 15a plots such a polygon for the 6th degree nonparametric curve used in previous examples. For n even $\mathbf{b}_q$ is the center Bézier point of $\mathbf{p}(t)$. For n odd, $\mathbf{b}_{q-1}^0, \mathbf{b}_{q-1}^1$ are the pair of interior Bézier points $\mathbf{b}_{q-1}, \mathbf{b}_q$. Hence, for the cases $n = \{1, 2, 3\}$, P coincides with the Bézier polygon.

The main advantage of this polygon over the standard Bézier polygon is that degree reduction is absolutely trivial: just “delete” inner control points. This operation yields Hermite interpolants of the curve, as shown in Figure 16 for the nonparametric curve of degree 6 used in the preceding example. If the central point $\mathbf{b}_3$ is removed, we get the polygon P_2 associated with the Hermite interpolant of order $k = 2$ (degree 5). If now the inner points $\mathbf{b}_2^{0/1}$ are deleted, we obtain the Hermite interpolant of order $k = 1$ (degree 3). Finally, after removing the pair $\mathbf{b}_1^{0/1}$ we get the Hermite interpolant of order $k = 0$, that is, a straight line connecting the curve endpoints. Degree

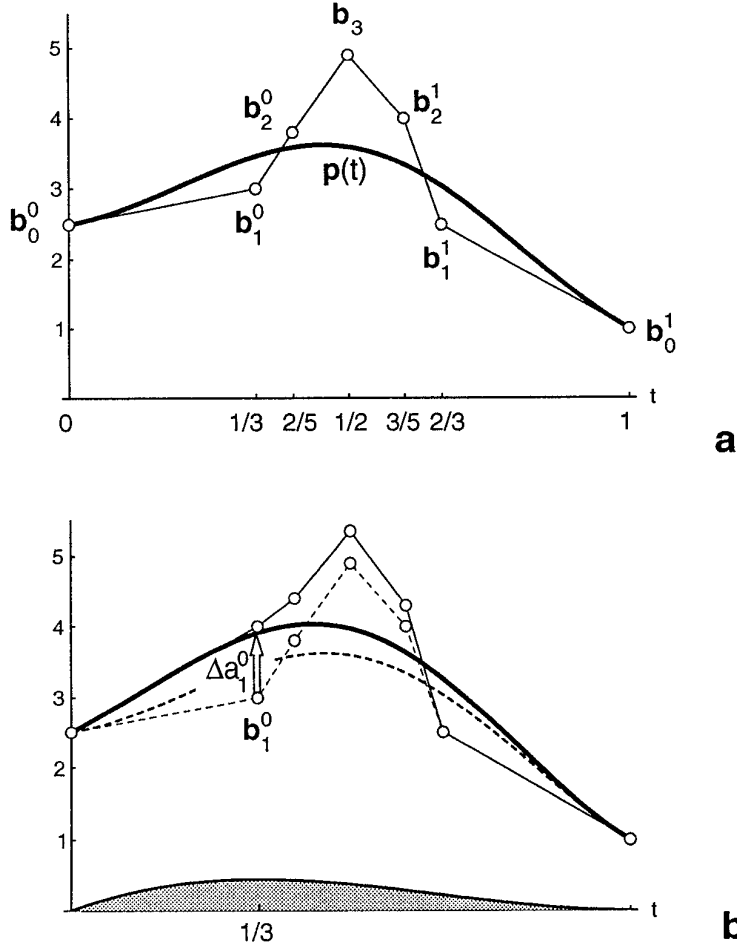


Fig. 15. (a) Control polygon P of a nonparametric curve ($n = 6$); (b) Effect of modifying a normalized coefficient.

elevation is performed by adding new points that, for high degrees, cluster around the central part of the curve. In contrast, using the Bézier representation all inner control points would change in a degree elevation/reduction process.

The Hermite interpolant $\mathbf{h}^j(t)$ is the truncated symmetric power form of degree $2j + 1$. Hence the expression for $\mathbf{b}_j^0$ as a function of the normalized coefficients is derived from the nonzero elements of the j th rows (22) in the conversion matrix $\mathbf{M}^{-1}$ (21), setting $n = 2j + 1$. By symmetry, the expression for $\mathbf{b}_j^0$ is computed swapping superscripts 0, 1:

$$\mathbf{b}_j^{0/1} = \mathbf{a}_j^{0/1} + \sum_{k=0}^{j-1} [\mathbf{a}_k^{0/1} \mathbf{M}_{j,k}^{-1} + \mathbf{a}_k^{1/0} \mathbf{M}_{n-j,k}^{-1}]. \quad (39)$$

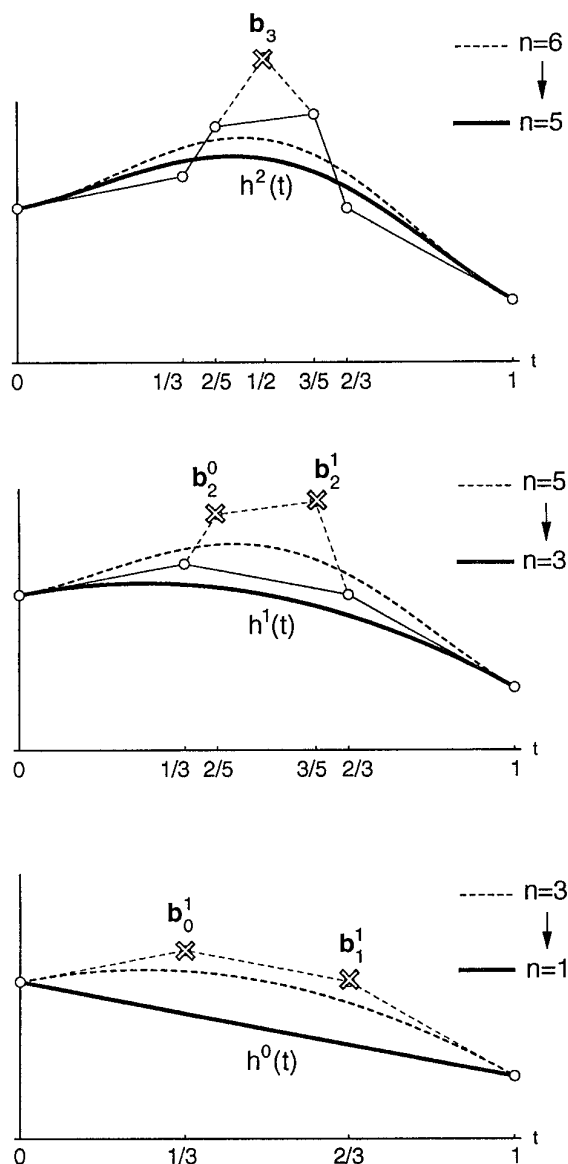


Fig. 16. Successive degree-reductions of a nonparametric curve.

The center point $\mathbf{b}_q$ (n even) is obtained from (33):

$$\mathbf{b}_q = \mathbf{a}_q + \frac{1}{2} (\mathbf{b}_{q-1}^0 + \mathbf{b}_{q-1}^1), \quad n = 2q.$$

A certain normalized coefficient of subscript j can be obtained as a linear combination of points of subscript $\leq j$ by recursively applying (39). Therefore the control polygon P uniquely defines a curve. Thus we can set its

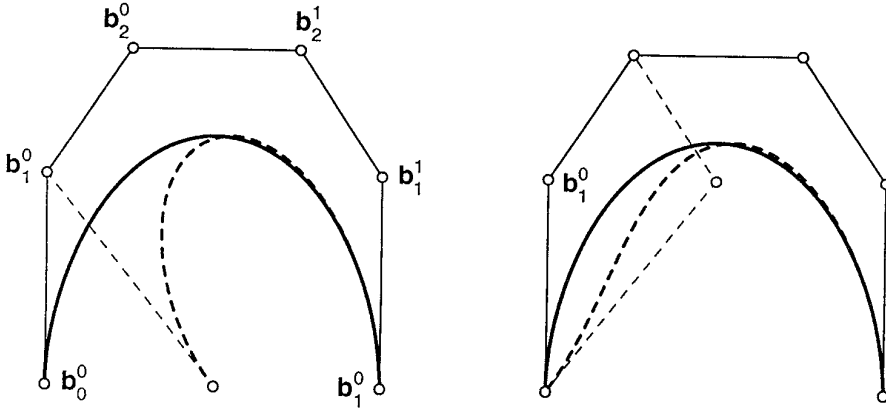


Fig. 17. Using the control points of polygon P as shape handles (degree $n = 5$).

points freely and use them as shape handles (Figure 17). Intuitively speaking, this polygon controls the curve by governing the shape of successive Hermite interpolants. This would be very difficult to achieve using the standard Hermite polynomials [Hoschek and Lasser 1993], because one must assign nonintuitive derivative magnitudes.

Similarly to the standard Bézier polygon, the control polygon P enjoys the following positive properties.

- The curve is tangent to P_1 at its endpoints, and in consequence to P .
- The curve is contained in the convex hull of P (See Appendix).
- Affine invariance, as both the Hermite interpolants and the Bézier form enjoy affine invariance.

Note that, if we modify a certain coefficient $\mathbf{a}_j^{0/1}$, then, in addition to $\mathbf{a}_j^{0/1}$, all inner points $\mathbf{b}_k^{0/1}$ with $k > j$ are affected, as derived from (39) and shown in Figure 15b ($j = 1$). Hence if we want to control independently each coefficient $\mathbf{a}_j^{0/1}$ through its associated control point $\mathbf{b}_j^{0/1}$, then moving $\mathbf{b}_j^{0/1}$ would imply moving all inner control points of subscript $k \geq j$. This fact suggests an alternative behavior of the polygon P , in which control points are not independent of each other, but follow a hierarchy. To illustrate this behavior, Figure 18 shows the effect of moving the control point $\mathbf{b}_0$ of a cubic curve. In this simple case, P coincides with the Bézier polygon, and there are two possible results of displacing such a control point. In the standard Bézier case the remaining control points are not altered, whereas, if this displacement is intended to modify just the associated symmetric power coefficient $\mathbf{a}_0^0$ and keep unaltered all other coefficients, then the inner control points are also moved. The idea of imposing a certain interdependence between control points of a cubic curve as they are manipulated is not new. Many commercial drawing applications [Computer Associates 1991; Claris 1994] implement a somewhat different relationship. Endpoints $\mathbf{b}_0, \mathbf{b}_3$ are called “anchor points,” and inner points

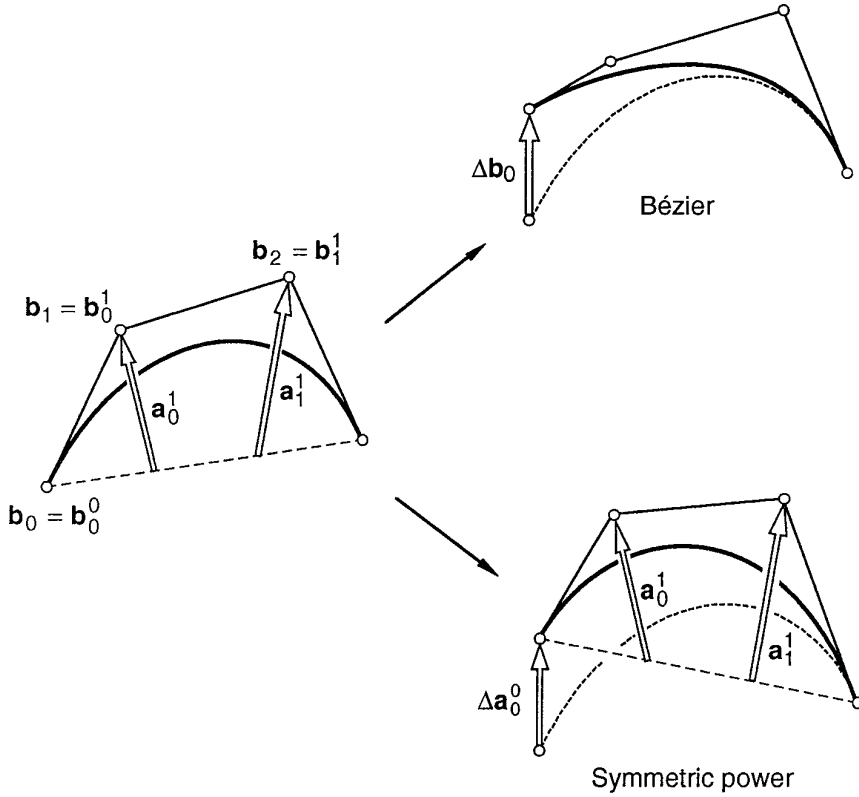


Fig. 18. Displacing a control point of a cubic curve.

$\mathbf{b}_1, \mathbf{b}_2$ are dubbed “direction points.” As one anchor point is moved (for instance $\mathbf{b}_0$), the attached direction point $\mathbf{b}_1$ suffers the same displacement.

5.5 Surfaces in the Symmetric Power Form

A bivariate rectangular surface $\mathbf{p}(u, v)$ of degree (m, n) over the domain $(u, v) \in [0, 1] \times [0, 1]$ is defined in the symmetric power form by using the customary tensor product approach. For the sake of simplicity, it is assumed both $(m = 2p - 1, n = 2q - 1)$ are odd degrees. Thus the surface can be decomposed into four terms:

$$\mathbf{p}(u, v) = \mathbf{p}^{(0,0)} + \mathbf{p}^{(1,0)} + \mathbf{p}^{(0,1)} + \mathbf{p}^{(1,1)}. \quad (40)$$

The first term,

$$\mathbf{p}^{(0,0)}(u, v) = \sum_{i=0}^{p-1} \sum_{j=0}^{q-1} \mathbf{a}_{i,j}^{(0,0)} P_i^0(u) P_j^0(v),$$

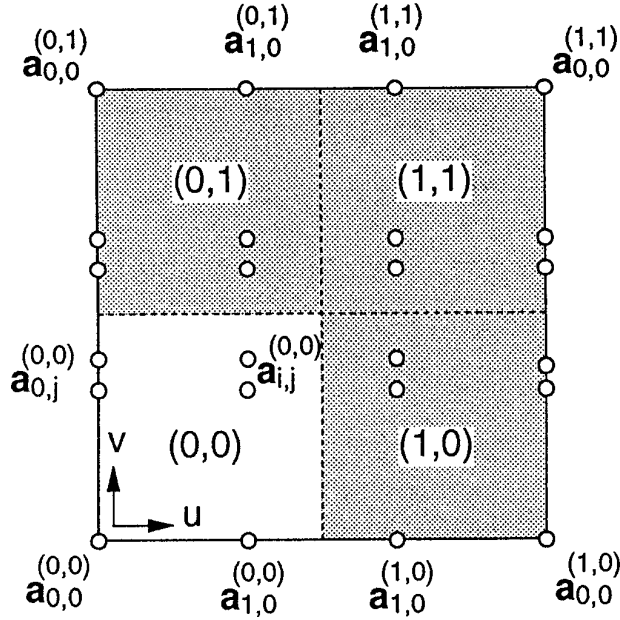


Fig. 19. Surface coefficients in the symmetric power form. Degree $(m, n) = (3, 5)$.

groups products of basis functions in u, v with superscripts 0, 0, respectively. The second term $\mathbf{p}^{(1,0)}(u, v)$ groups those with superscripts 1, 0 and so on. The basis functions in $\mathbf{p}^{(0,0)}(u, v)$ attain maxima at

$$(u_i, v_j)^{(0,0)} = \left(\frac{i}{2i+1}, \frac{j}{2j+1} \right), \quad \begin{matrix} i = 0, \dots, p-1 \\ j = 0, \dots, q-1, \end{matrix} \quad (41)$$

that is, within the quarter $[0, 1/2) \times [0, 1/2)$ containing the corner $(u, v) = (0, 0)$. In consequence, the corresponding coefficients $\mathbf{a}_{i,j}^{(0,0)}$ control the shape of the surface mostly in such a quarter of the parametric domain. Analogously, the superscripts of the remaining terms in (40) indicate the coordinates of the corner corresponding to the quarter of predominant influence.

Given the shape of the basis functions, the surface coefficients can be interpreted as follows (see Figure 19).

- Coefficients subscripted $_{0,0}$, that is, $\mathbf{a}_{0,0}^{(0,0)}$, $\mathbf{a}_{0,0}^{(0,1)}$, $\mathbf{a}_{0,0}^{(1,0)}$, $\mathbf{a}_{0,0}^{(1,1)}$, are the corner points of the surface. The superscript indicates the corner coordinates.
- A set of coefficients with a null subscript determines a boundary curve. For instance, the boundary $u = 0$ is determined by $\{\mathbf{a}_{0,j}^{(0,0)}, \mathbf{a}_{0,j}^{(0,1)}\}_{j=0}^{q-1}$.
- If all inner coefficients (those with no null subscript) are zero, we obtain the bilinearly blended Coons patch that interpolates the four boundaries. Therefore these interior coefficients are vectors expressing how much the surface differs from the Coons patch.

Similarly to the univariate case, degree elevation is performed just by setting new nonzero coefficients. This is much simpler than in the Bézier basis, for which all coefficients change as all rows or columns of control points must be degree-elevated. Degree reduction is also easier and much more flexible than in the Bézier case. All bivariate Bernstein polynomials have degree (m, n) , and therefore degree reduction is usually performed for all rows of control points and then for the resulting columns [Eck 1993]. In the symmetric power case, the basis functions have different degrees, and degree reduction reduces to canceling the desired high degree terms. Now we can choose individually which terms are eliminated, something impossible to achieve in the Bézier form.

Figure 20(a) shows a nonparametric surface of degree $(m, n) = (3, 5)$, whose coefficients $\mathbf{a}_{i,j}^{(0,0)}$ are represented by the length of bars raised at corresponding maxima (41), and analogously for other coefficients. In Figure 20(b) this surface is degree-reduced without modifying the boundaries, by canceling those inner coefficients of subscript $(i, j) = (1, 2)$. Finally, as already stated, if all the inner coefficients are canceled (Figure 20(c)) one gets a bilinearly blended Coons patch, which is assumed to be the smoothest surface that interpolates four prescribed boundaries [Brunet 1985; Farin 1993]. In the new basis this assumption becomes evident, because for boundaries of degree (m, n) the nonvanishing surface components have degrees at most $(m, 1)$ or $(1, n)$, and high-degree terms are associated with unwanted oscillations. Thus we can think of the degree-reduction process as a smoothing procedure as well.

6. CONCLUSIONS

The symmetric power form has been derived as the proper symmetric analogue over $t \in [0, 1]$ of the power form. This new form can be regarded as an Hermite two-point expansion, instead of a Taylor expansion. Similarly to the power form, it can be evaluated using a fast Horner-like scheme. The transformation between the Bernstein and symmetric power forms is not ill-conditioned. As a by-product, this transformation performs a degree reduction if the Bézier polynomial has a degree higher than necessary, which speeds up the evaluation. As for numerical properties, this form is much closer to the Bernstein form than to the monomial form, avoiding undesirable gaps between consecutive segments in the piecewise case.

The components of a polynomial in the symmetric power form can be interpreted as the neglected terms after performing successive degree reductions following Forrest's algorithm. Multiplying the symmetric power polynomials by binomial factors, normalized functions are obtained, which are pairs of central Bernstein polynomials for successive increasing degrees. The corresponding normalized coefficients admit a geometric interpretation related to nonparametric Bézier polygons of successive Hermite interpolants of the polynomial, and define a polygon that provides a way to control the shape of a curve. For bivariate surfaces, certain coefficients

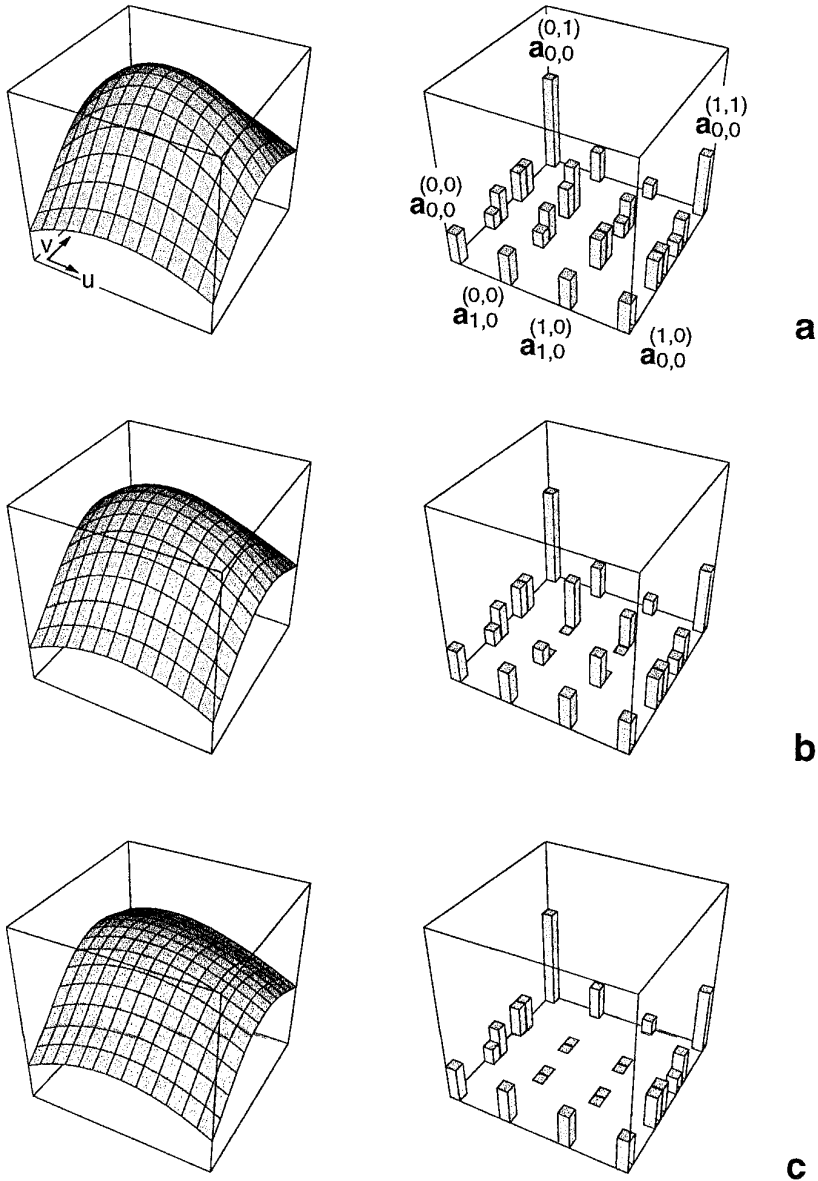


Fig. 20. Effect of canceling inner surface coefficients. Degree $(m, n) = (3, 5)$.

determine the boundary curves, and the remaining ones express the deviation between the surface and the bilinearly blended Coons patch interpolating the boundaries.

The symmetric power basis functions have increasing degrees, a property not shared by the Bernstein polynomials. A polynomial in the new form is thus decomposed into linear, cubic, quintic, and successive components. Hence we can control these individual components corresponding to differ-

ent degrees in a way that is impossible to achieve in the Bézier form, so that, for instance, degree elevation and reduction become trivial. In addition, by inspecting the coefficients, it is always detected without numerical error whether a curve is a straight line. We conclude that the symmetric power form is better suited to general degree raising and lowering than the Bézier representation, and provides an efficient environment that deals gracefully with polynomials of different degrees.

APPENDIX. CONTAINMENT IN THE CONVEX HULL

In this Appendix it is proved that the Bézier points $\{\mathbf{b}_i\}$ of a curve $\mathbf{p}(t)$ lie in the convex hull of the polygon P (38) and, in consequence, $\mathbf{p}(t)$ too. This result implies that in general the Bézier polygon hugs the curve tighter. The demonstration proceeds by induction.

- (1) Assume that the Bézier points of the Hermite interpolant $\mathbf{h}^{k-1}(t)$ are contained in the convex hull of the polygon P_{k-1} . Those of its degree-elevated form $\hat{\mathbf{h}}^{k-1}(t)$ are contained too, because the degree elevation is a corner-cutting procedure. According to (34), the Bézier points of $\mathbf{h}^k(t)$ are those of $\hat{\mathbf{h}}^{k-1}(t)$, except that two points are replaced by $\mathbf{b}_k^0, \mathbf{b}_k^1$. It is concluded that the Bézier points of $\hat{\mathbf{h}}^{k-1}(t)$ are contained in the convex hull of the new polygon $P_k = \{P_{k-1}, \mathbf{b}_k^0, \mathbf{b}_k^1\}$ (37).
- (2) The preceding assumption holds true for $k = 1$: $\mathbf{h}^0(t)$ (the straight line segment $\mathbf{a}_0^0, \mathbf{a}_0^1$) is contained in the convex hull of $P_0 = \{\mathbf{a}_0^0, \mathbf{a}_0^1\} = \{\mathbf{b}_0^0, \mathbf{b}_0^1\}$.
- (3) Apply the preceding reasoning for $k = 1, \dots, q-1$ and reach P_{q-1} . It is concluded that the Bézier points of $\mathbf{h}^{q-1}(t)$ lie in the convex hull of P_{q-1} .
- (4) For n odd, $P = P_{q-1}$ and the Bézier points $\{\mathbf{b}_i\}$ of $\mathbf{p}(t)$ are those of $\mathbf{h}_{q-1}(t)$. Therefore they lie in the convex hull of P .
- (5) For n even, Equation (33) indicates that the Bézier points $\{\mathbf{b}_i\}$ of $\mathbf{p}(t)$ are those of the degree-elevated $\mathbf{h}^{q-1}(t)$, except $\mathbf{b}_q$. It is concluded that $\{\mathbf{b}_i\}$ are contained within the convex hull of $P = \{P_{q-1}, \mathbf{b}_q\}$.

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